MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
49PR POWER PLANT PIPE RACK
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
STAAD	:	Commonly used analysis and design software package named STAAD.Pro
Sec.	:	Section
max.	:	Maximum
ref.	:	Reference
No.	:	Number
cl.	:	Clause
nos.	:	Numbers
Col.	:	Column
approx.	:	Approximate
eqn.	:	Equation
mkd.	:	Marked
Dia.	:	Diameter
Spac.	:	Spacing
Rebar	:	Reinforcement Bar

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1. INTENT

The purpose of this document is to analysis and design of foundations for 49PR power plant on ground pipe rack cum cable rack.The foundation sizes are calculated to ensure structural integrity and safety for the design load combinations. The analysis and design is carried out in accordance with project specifications, codes and standards.
Foundation for ground pipe supports GPS-6 & GPS-7 are covered in this design document.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and STAAD. Load are taken from super structure STAAD file of GPS-1 & GPS-5. Sliding & overturning check are done in excel sheet. STAAD modelling has been done with plate elements.
Modulus of Subgrade reaction of 54000 kN/m^2/m and net allowable bearing capacity of 450 kN/m^2 have been considered as per soil report.

4. DESIGN PARAMETER:

- Concrete Strength of 28 days : For foundation & pedestals = 35MPa
  Binding concrete below foundation base = 15 MPa
- Reinforcement Grade : Deformed Bar : ASTM A 615/ A 615M Grade 60 (fy= 420MPa)
  Welded Wire Fabric : ASTM A 185 (Plain)/ A 497 (Deformed)
- Grouting : Grout shall be 30 mm cementitious of Strength 40 Mpa

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5. PRIMARY LOAD CASES CONSIDERED :

Primary Load Case	Description	Direction	Remarks	STAAD Nomenclature	Load Case Number
Dead Load	-	Downward	-	D	1
Live Load	-	Downward	-	L	2
	-	Downward	-	L1	3
Wind Load	-	Along +Z	-	WL (+Z)	11
	-	Along -Z	-	WL (-Z)	12
	-	Along +X	-	WL (+X)	13
	-	Along -X	-	WL (-X)	14
Earthquake Load	-	Along +Z	-	EL (+Z)	30
	-	Along -Z	-	EL (-Z)	31
	-	Along +X	-	EL (+X)	32
	-	Along -X	-	EL (-X)	33
Temperature Load	-	-	FOR +25°C	TL1	20
	-	-	FOR -25°C	TL2	21

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6. LOAD COMBINATIONS CONSIDERED:

Load combinations are determined accordance with SBC 301, SBC 304 and Civil, Structural Design Criteria document number MD-216-1000-EG-ST-CRT-1001.

Factored Load Combinations for strength design:

[ref. cl. 2.3.2 : SBC 301-2007]

Serial No.	STAAD Load Combination Number	Description
1	100	1.4xD
2	101	1.2xD + 1.2xTL1 + 1.6xL
3	102	1.2xD + 1.2xTL1 + 1.6xL1
4	103	1.2xD + 1.2xTL2 + 1.6xL
5	104	1.2xD + 1.2xTL2 + 1.6xL1
6	105	1.2xD + 0.8xWL(+Z)
7	106	1.2xD + 0.8xWL(+X)
8	107	1.2xD + 0.8xWL(-Z)
9	108	1.2xD + 0.8xWL(-X)
10	109	0.9xD + 1.6xWL(+Z)
11	110	0.9xD + 1.6xWL(+X)
12	111	0.9xD + 1.6xWL(-Z)
13	112	0.9xD + 1.6xWL(-X)
14	113	1.2xD + 0.5xL + 1.6xWL (+Z)
15	114	1.2xD + 0.5xL + 1.6xWL (+X)
16	115	1.2xD + 0.5xL + 1.6xWL (-Z)
17	116	1.2xD + 0.5xL + 1.6xWL (-X)
18	117	1.2xD + 0.5xL1 + 1.6xWL (+Z)
19	127	1.2xD + 0.5xL1 + 1.6xWL (+X)
20	118	1.2xD + 0.5xL1 + 1.6xWL (-Z)
21	119	1.2xD + 0.5xL1 + 1.6xWL (-X)
22	128	1.2xD + 0.5xL + 1xEL (+Z)
23	120	1.2xD + 0.5xL + 1xEL (+X)
24	121	1.2xD + 0.5xL + 1xEL (-Z)
25	122	1.2xD + 0.5xL + 1xEL (-X)
26	129	1.2xD + 0.5xL1 + 1xEL (+Z)
27	123	1.2xD + 0.5xL1 + 1xEL (+X)
28	124	1.2xD + 0.5xL1 + 1xEL (-Z)
29	125	1.2xD + 0.5xL1 + 1xEL (-X)
30	126	0.9xD + 1xEL(+Z)
31	130	0.9xD + 1xEL(+X)
32	131	0.9xD + 1xEL(-Z)
33	132	0.9xD + 1xEL(-X)

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Factored Load Combinations for serviceability design:

[ref. cl. 2.4.1 : SBC 301-2007]

Serial No.	STAAD Load Combination Number	Description
1	201	D + L + TL1
2	202	D + L1 + TL1
3	203	D + L + TL2
4	204	D + L1 + TL2
5	205	D + 0.75xL + 0.75xTL1
6	206	D + 0.75xL1 + 0.75xTL1
7	207	D + 0.75xL + 0.75xTL2
8	208	D + 0.75xL1 + 0.75xTL2
9	209	D + WL(+Z)
10	210	D + WL(+X)
11	211	D + WL(-Z)
12	212	D + WL(-X)
13	213	D + 0.7xEL(+Z)
14	214	D + 0.7xEL(+X)
15	215	D + 0.7xEL(-Z)
16	216	D + 0.7xEL(-X)
17	217	D + 0.75xL + 0.75xWL(+Z)
18	218	D + 0.75xL1 + 0.75xWL(+Z)
19	219	D + 0.75xL + 0.75xWL(-Z)
20	220	D + 0.75xL1 + 0.75xWL(-Z)
21	221	D + 0.75xL + 0.75xWL(+X)
22	222	D + 0.75xL1 + 0.75xWL(+X)
23	223	D + 0.75xL + 0.75xWL(-X)
24	224	D + 0.75xL1 + 0.75xWL(-X)
25	225	D + 0.75xL + 0.525xEL(+Z)
26	226	D + 0.75xL1 + 0.525xEL(+Z)
27	227	D + 0.75xL + 0.525xEL(+X)
28	228	D + 0.75xL1 + 0.525xEL(+X)
29	229	D + 0.75xL + 0.525xEL(-Z)
30	230	D + 0.75xL1 + 0.525xEL(-Z)
31	231	D + 0.75xL + 0.525xEL(-X)
32	232	D + 0.75xL1 + 0.525xEL(-X)
33	233	0.6xD + WL(+Z)
34	234	0.6xD + WL(+X)
35	235	0.6xD + WL(-Z)
36	236	0.6xD + WL(-X)
37	237	0.6xD + 0.7xEL(+Z)
38	238	0.6xD + 0.7xEL(+X)
39	239	0.6xD + 0.7xEL(-Z)
40	240	0.6xD + 0.7xEL(-X)

7. ALLOWABLE SOIL BEARING CAPACITY:

Net safe bearing capacity of soil considered as 450kN/m2 and vertical modulus of subgrade reaction considered as 54000 kN/m2/mat at 1.5m below EGL as per soil report. [refer MD-216-3000-EG-CV-RPT-1008].

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8. DESIGN CALCULATION OF FOOTINGS

• DESIGN CALCULATION OF FOOTING MKD. GPS-6/ F1:

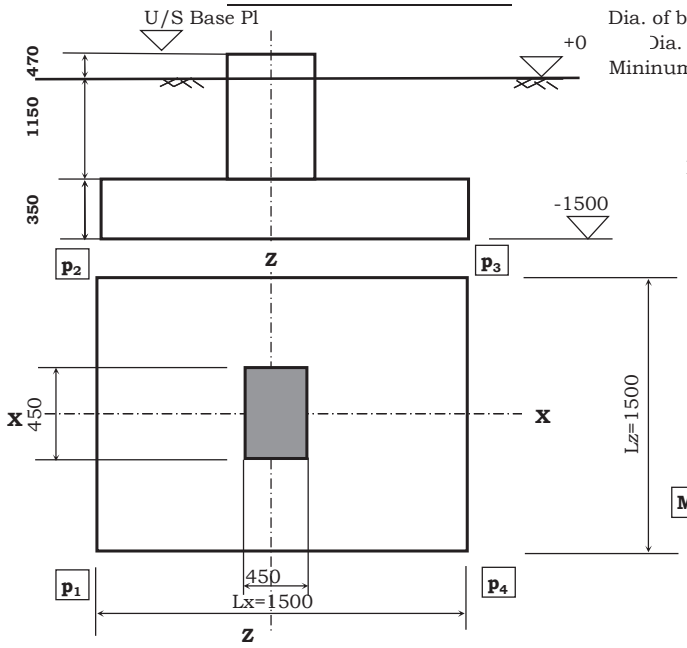
	Along-X	Along-Z
Size of column :	0.45m	0.45m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-1.50	m
U/S level Base Plate :	0.47	m
Unit weight of conc :	24	kN/m³
Unit weight of soil :	18	kN/m³

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.5*1.5*0.35*24)+(0.45*0.45*1.62*24	26.77KN
Weight of soil =	((1.5*1.5)-(0.45*0.45))*1.15*18 =	42.38KN
	Total =	69.16KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

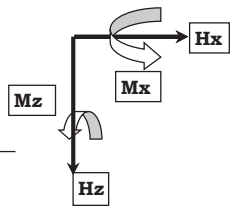


Effective depth of slab along X direction = 292mm = 0.292m

Effective depth of slab along Z direction = 276mm = 0.276m

Clear projection along Z direction = 0.525m

Clear projection along X direction = 0.525m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sensitivity: LNT Construction Internal Use

• DESIGN CALCULATION OF FOOTING MKD. GPS-6/ F1:

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 D + L + TL1	11.1	-1.3	1.3	-1.2	0.0	80.23	-1.27	1.32	-1.19	0.00
202 D + L1 + TL1	8.5	-1.3	1.3	1.3	0.0	77.66	-1.27	1.32	1.35	0.00
203 D + L + TL2	11.1	-1.3	1.3	-1.4	0.0	80.23	-1.27	1.32	-1.35	0.00
204 D + L1 + TL2	8.5	-1.3	1.3	1.2	0.0	77.66	-1.27	1.32	1.18	0.00
205 D + 0.75XL + 0.75XTL1	8.6	-1.0	1.0	-0.9	0.0	77.78	-0.95	0.99	-0.89	0.00
206 D + 0.75XL1 + 0.75XTL1	6.7	-1.0	1.0	1.0	0.0	75.85	-0.95	0.99	1.01	0.00
207 D + 0.75XL + 0.75XTL2	8.6	-1.0	1.0	-1.0	0.0	77.78	-0.95	0.99	-1.02	0.00
208 D + 0.75XL1 + 0.75XTL2	6.7	-1.0	1.0	0.9	0.0	75.85	-0.95	0.99	0.89	0.00
209 D + WL(+Z)	4.2	0.0	0.0	-2.8	0.0	73.37	0.00	0.00	-2.75	0.00
210 D + WL(+X)	1.3	-2.8	3.0	0.0	0.0	70.42	-2.75	3.03	0.00	0.00
211 D + WL(-Z)	-1.7	0.0	0.0	2.8	0.0	67.47	0.00	0.00	2.75	0.00
212 D + WL(-X)	1.3	2.8	-3.0	0.0	0.0	70.42	2.75	-3.03	0.00	0.00
213 D + 0.7XEL(+Z)	1.5	0.0	0.0	-0.2	0.0	70.65	0.00	0.00	-0.15	0.00
214 D + 0.7XEL(+X)	1.3	-0.2	0.2	0.0	0.0	70.42	-0.15	0.23	0.00	0.00
215 D + 0.7XEL(-Z)	1.0	0.0	0.0	0.2	0.0	70.20	0.00	0.00	0.15	0.00
216 D + 0.7XEL(-X)	1.3	0.2	-0.2	0.0	0.0	70.42	0.15	-0.23	0.00	0.00
217 D + 0.75XL + 0.75XWL(+Z)	10.8	-1.0	1.0	-3.0	0.0	80.00	-0.95	0.99	-3.02	0.00
218 D + 0.75XL1 + 0.75XWL(+Z)	8.9	-1.0	1.0	-1.1	0.0	78.06	-0.95	0.99	-1.11	0.00
219 D + 0.75XL + 0.75XWL(-Z)	6.4	-1.0	1.0	1.1	0.0	75.57	-0.95	0.99	1.11	0.00
220 D + 0.75XL1 + 0.75XWL(-Z)	4.5	-1.0	1.0	3.0	0.0	73.63	-0.95	0.99	3.01	0.00
221 D + 0.75XL + 0.75XWL(+X)	8.6	-3.0	3.3	-1.0	0.0	77.78	-3.01	3.26	-0.96	0.00
222 D + 0.75XL1 + 0.75XWL(+X)	6.7	-3.0	3.3	0.9	0.0	75.85	-3.01	3.26	0.95	0.00
223 D + 0.75XL + 0.75XWL(-X)	8.6	1.1	-1.3	-1.0	0.0	77.78	1.11	-1.28	-0.96	0.00
224 D + 0.75XL1 + 0.75XWL(-X)	6.7	1.1	-1.3	0.9	0.0	75.85	1.11	-1.28	0.95	0.00
225 D + 0.75XL + 0.525XEL(+Z)	8.8	-1.0	1.0	-1.1	0.0	77.95	-0.95	0.99	-1.07	0.00
226 D + 0.75XL1 + 0.525XEL(+Z)	6.9	-1.0	1.0	0.8	0.0	76.02	-0.95	0.99	0.83	0.00
227 D + 0.75XL + 0.525XEL(+X)	8.6	-1.1	1.2	-1.0	0.0	77.78	-1.07	1.16	-0.96	0.00
228 D + 0.75XL1 + 0.525XEL(+X)	6.7	-1.1	1.2	0.9	0.0	75.85	-1.07	1.16	0.95	0.00
229 D + 0.75XL + 0.525XEL(-Z)	8.5	-1.0	1.0	-0.8	0.0	77.61	-0.95	0.99	-0.84	0.00
230 D + 0.75XL1 + 0.525XEL(-Z)	6.5	-1.0	1.0	1.1	0.0	75.68	-0.95	0.99	1.06	0.00
231 D + 0.75XL + 0.525XEL(-X)	8.6	-0.8	0.8	-1.0	0.0	77.78	-0.84	0.82	-0.96	0.00
232 D + 0.75XL1 + 0.525XEL(-X)	6.7	-0.8	0.8	0.9	0.0	75.85	-0.84	0.82	0.95	0.00
233 0.6XD + WL(+Z)	3.7	0.0	0.0	-2.8	0.0	72.87	0.00	0.00	-2.75	0.00
234 0.6XD + WL(+X)	0.8	-2.8	3.0	0.0	0.0	69.92	-2.75	3.03	0.00	0.00
235 0.6XD + WL(-Z)	-2.2	0.0	0.0	2.8	0.0	66.97	0.00	0.00	2.75	0.00
236 0.6XD + WL(-X)	0.8	2.8	-3.0	0.0	0.0	69.92	2.75	-3.03	0.00	0.00
237 0.6XD + 0.7XEL(+Z)	1.0	0.0	0.0	-0.2	0.0	70.14	0.00	0.00	-0.15	0.00
238 0.6XD + 0.7XEL(+X)	0.8	-0.2	0.2	0.0	0.0	69.92	-0.15	0.23	0.00	0.00
239 0.6XD + 0.7XEL(-Z)	0.5	0.0	0.0	0.2	0.0	69.69	0.00	0.00	0.15	0.00
240 0.6XD + 0.7XEL(-X)	0.8	0.2	-0.2	0.0	0.0	69.92	0.15	-0.23	0.00	0.00

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Sensitivity: LNT Construction Internal Use

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
100 1.4XD	1.8	0.0	0.0	0.0	0.0	70.93	0.00	0.00	0.00	0.00
101 1.2XD + 1.2XTL1 + 1.6XL	17.2	-2.0	2.1	-1.9	0.0	86.37	-2.03	2.11	-1.94	0.00
102 1.2XD + 1.2XTL1 + 1.6XL1	13.1	-2.0	2.1	2.1	0.0	82.25	-2.03	2.11	2.12	0.00
103 1.2XD + 1.2XTL2 + 1.6XL	17.2	-2.0	2.1	-2.1	0.0	86.37	-2.03	2.11	-2.13	0.00
104 1.2XD + 1.2XTL2 + 1.6XL1	13.1	-2.0	2.1	1.9	0.0	82.25	-2.03	2.11	1.93	0.00
105 1.2XD + 0.8XWL(+Z)	3.9	0.0	0.0	-2.2	0.0	73.04	0.00	0.00	-2.20	0.00
106 1.2XD + 0.8XWL(+X)	1.5	-2.2	2.4	0.0	0.0	70.68	-2.20	2.42	0.00	0.00
107 1.2XD + 0.8XWL(-Z)	-0.8	0.0	0.0	2.2	0.0	68.31	0.00	0.00	2.20	0.00
108 1.2XD + 0.8XWL(-X)	1.5	2.2	-2.4	0.0	0.0	70.68	2.20	-2.42	0.00	0.00
109 0.9XD + 1.6XWL(+Z)	5.9	0.0	0.0	-4.4	0.0	75.02	0.00	0.00	-4.40	0.00
110 0.9XD + 1.6XWL(+X)	1.1	-4.4	4.8	0.0	0.0	70.30	-4.40	4.84	0.00	0.00
111 0.9XD + 1.6XWL(-Z)	-3.6	0.0	0.0	4.4	0.0	65.57	0.00	0.00	4.40	0.00
112 0.9XD + 1.6XWL(-X)	1.1	4.4	-4.8	0.0	0.0	70.30	4.40	-4.84	0.00	0.00
113 1.2XD + 0.5XL + 1.6XWL (+Z)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
114 1.2XD + 0.5XL + 1.6XWL (+X)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
115 1.2XD + 0.5XL + 1.6XWL (-Z)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
116 1.2XD + 0.5XL + 1.6XWL (-X)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
117 1.2XD + 0.5XL1 + 1.6XWL (+Z)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
118 1.2XD + 0.5XL1 + 1.6XWL (+X)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
121 1.2XD + 0.5XL + EL (+Z)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
122 1.2XD + 0.5XL + EL (+X)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
123 1.2XD + 0.5XL + EL (-Z)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
124 1.2XD + 0.5XL + EL (-X)	6.4	-0.6	0.7	-0.6	0.0	75.58	-0.64	0.66	-0.64	0.00
125 1.2XD + 0.5XL1 + EL (+Z)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
126 1.2XD + 0.5XL1 + EL (+X)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
127 1.2XD + 0.5XL1 + EL (-Z)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
128 1.2XD + 0.5XL1 + EL (-X)	5.1	-0.6	0.7	0.6	0.0	74.29	-0.64	0.66	0.63	0.00
129 0.9XD + EL(+Z)	1.5	0.0	0.0	-0.2	0.0	70.62	0.00	0.00	-0.22	0.00
130 0.9XD + EL(+X)	1.1	-0.2	0.3	0.0	0.0	70.30	-0.22	0.33	0.00	0.00
131 0.9XD + EL(-Z)	0.8	0.0	0.0	0.2	0.0	69.98	0.00	0.00	0.22	0.00
132 0.9XD + EL(-X)	1.1	0.2	-0.3	0.0	0.0	70.30	0.22	-0.33	0.00	0.00

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Sensitivity: LNT Construction Internal Use

Check for overturning and sliding:

Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning F.O.S. Mz	Overturning F.O.S. Mx	Sliding F.O.S. Hx	Sliding F.O.S. Hz
201 D + L + TL1	80.2	-1.3	1.3	-1.2	0.0	45.6	NA	25.3	26.9
202 D + L1 + TL1	77.7	-1.3	1.3	1.3	0.0	44.1	NA	24.5	23.1
203 D + L + TL2	80.2	-8.0	1.3	-1.4	0.0	45.6	NA	4.0	23.7
204 D + L1 + TL2	77.7	-1.3	1.3	1.2	0.0	44.1	NA	24.5	26.2
225 D + 0.75XL + 0.525XEL(+Z)	77.9	-1.0	1.0	-1.1	0.0	59.0	NA	32.8	29.1
226 D + 0.75XL1 + 0.525XEL(+Z)	76.0	-1.0	1.0	0.8	0.0	57.5	NA	31.9	36.5
227 D + 0.75XL + 0.525XEL(+X)	77.8	-1.1	1.2	-1.0	0.0	50.2	NA	29.2	32.6
228 D + 0.75XL1 + 0.525XEL(+X)	75.8	-1.1	1.2	0.9	0.0	48.9	NA	28.4	32.0
229 D + 0.75XL + 0.525XEL(-Z)	77.6	-1.0	1.0	-0.8	0.0	58.7	NA	32.6	37.0
230 D + 0.75XL1 + 0.525XEL(-Z)	75.7	-1.0	1.0	1.1	0.0	57.3	NA	31.8	28.5
231 D + 0.75XL + 0.525XEL(-X)	77.8	-0.8	0.8	-1.0	0.0	71.2	NA	37.2	32.6
232 D + 0.75XL1 + 0.525XEL(-X)	75.8	-0.8	0.8	0.9	0.0	69.5	NA	36.2	32.0
233 0.6XD + WL(+Z)	72.9	0.0	0.0	-2.8	0.0	NA	NA	NA	10.6
234 0.6XD + WL(+X)	69.9	-2.8	3.0	0.0	0.0	17.3	NA	10.2	NA
235 0.6XD + WL(-Z)	67.0	0.0	0.0	2.8	0.0	NA	NA	NA	9.7
236 0.6XD + WL(-X)	69.9	2.8	-3.0	0.0	0.0	17.3	NA	10.2	NA
237 0.6XD + 0.7XEL(+Z)	70.1	0.0	0.0	-0.2	0.0	NA	NA	NA	182.2
238 0.6XD + 0.7XEL(+X)	69.9	-0.2	0.2	0.0	0.0	228.0	NA	181.6	NA
239 0.6XD + 0.7XEL(-Z)	69.7	0.0	0.0	0.2	0.0	NA	NA	NA	182.2
240 0.6XD + 0.7XEL(-X)	69.9	0.2	-0.2	0.0	0.0	228.0	NA	181.6	NA
					Min.=	17.3	NA	4.0	9.7
						OK	OK	OK	OK

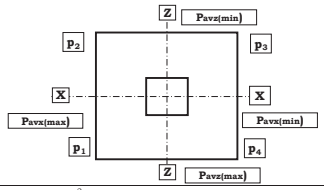
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Sensitivity: LNT Construction Internal Use

• DESIGN CALCULATION OF FOOTING MKD. GPS-6/ F1:



Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²

Allowable Stress Design

Load Combination	Design Load/moments			Base Pressure (kN/m²)												
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201 D + L + TL1	80.23	1.32	0.00	35.66	2.35	0.00	38.01	38.01	33.31	33.31	11.01	38.01	33.31	35.66	35.66	O.K
202 D + L1 + TL1	77.66	1.32	0.00	34.51	2.35	0.00	36.86	36.86	32.16	32.16	9.86	36.86	32.16	34.51	34.51	O.K
203 D + L + TL2	80.23	1.32	0.00	35.66	2.35	0.00	38.01	38.01	33.31	33.31	11.01	38.01	33.31	35.66	35.66	O.K
204 D + L1 + TL2	77.66	1.32	0.00	34.51	2.35	0.00	36.86	36.86	32.16	32.16	9.86	36.86	32.16	34.51	34.51	O.K
205 D + 0.75XL + 0.75XTL1	77.78	0.99	0.00	34.57	1.76	0.00	36.33	36.33	32.81	32.81	9.33	36.33	32.81	34.57	34.57	O.K
206 D + 0.75XL1 + 0.75XTL1	75.85	0.99	0.00	33.71	1.76	0.00	35.47	35.47	31.95	31.95	8.47	35.47	31.95	33.71	33.71	O.K
207 D + 0.75XL + 0.75XTL2	77.78	0.99	0.00	34.57	1.76	0.00	36.33	36.33	32.81	32.81	9.33	36.33	32.81	34.57	34.57	O.K
208 D + 0.75XL1 + 0.75XTL2	75.85	0.99	0.00	33.71	1.76	0.00	35.47	35.47	31.95	31.95	8.47	35.47	31.95	33.71	33.71	O.K
209 D + WL(+Z)	73.37	0.00	0.00	32.61	0.00	0.00	32.61	32.61	32.61	32.61	5.61	32.61	32.61	32.61	32.61	O.K
210 D + WL(+X)	70.42	3.03	0.00	31.30	5.38	0.00	36.68	36.68	25.92	25.92	9.68	36.68	25.92	31.30	31.30	O.K
211 D + WL(-Z)	67.47	0.00	0.00	29.99	0.00	0.00	29.99	29.99	29.99	29.99	2.99	29.99	29.99	29.99	29.99	O.K
212 D + WL(-X)	70.42	-3.03	0.00	31.30	5.38	0.00	36.68	36.68	25.92	25.92	9.68	36.68	25.92	31.30	31.30	O.K
213 D + 0.7XEL(+Z)	70.65	0.00	0.00	31.40	0.00	0.00	31.40	31.40	31.40	31.40	4.40	31.40	31.40	31.40	31.40	O.K
214 D + 0.7XEL(+X)	70.42	0.23	0.00	31.30	0.41	0.00	31.71	31.71	30.89	30.89	4.71	31.71	30.89	31.30	31.30	O.K
215 D + 0.7XEL(-Z)	70.20	0.00	0.00	31.20	0.00	0.00	31.20	31.20	31.20	31.20	4.20	31.20	31.20	31.20	31.20	O.K
216 D + 0.7XEL(-X)	70.42	-0.23	0.00	31.30	0.41	0.00	31.71	31.71	30.89	30.89	4.71	31.71	30.89	31.30	31.30	O.K
217 D + 0.75XL + 0.75XWL(+Z)	80.00	0.99	0.00	35.55	1.76	0.00	37.32	37.32	33.79	33.79	10.32	37.32	33.79	35.56	35.56	O.K
218 D + 0.75XL1 + 0.75XWL(+Z)	78.06	0.99	0.00	34.69	1.76	0.00	36.46	36.46	32.93	32.93	9.46	36.46	32.93	34.70	34.70	O.K
219 D + 0.75XL + 0.75XWL(-Z)	75.57	0.99	0.00	33.59	1.76	0.00	35.35	35.35	31.82	31.82	8.35	35.35	31.82	33.59	33.59	O.K
220 D + 0.75XL1 + 0.75XWL(-Z)	73.63	0.99	0.00	32.73	1.76	0.00	34.49	34.49	30.96	30.96	7.49	34.49	30.96	32.73	32.73	O.K
221 D + 0.75XL + 0.75XWL(+X)	77.78	3.26	0.00	34.57	5.80	0.00	40.36	40.36	28.77	28.77	13.36	40.36	28.77	34.57	34.57	O.K
222 D + 0.75XL1 + 0.75XWL(+X)	75.85	3.26	0.00	33.71	5.80	0.00	39.51	39.51	27.91	27.91	12.51	39.51	27.91	33.71	33.71	O.K
223 D + 0.75XL + 0.75XWL(-X)	77.78	-1.28	0.00	34.57	2.27	0.00	36.84	36.84	32.30	32.30	9.84	36.84	32.30	34.57	34.57	O.K
224 D + 0.75XL1 + 0.75XWL(-X)	75.85	-1.28	0.00	33.71	2.27	0.00	35.98	35.98	31.44	31.44	8.98	35.98	31.44	33.71	33.71	O.K
225 D + 0.75XL + 0.525XEL(+Z)	77.95	0.99	0.00	34.64	1.76	0.00	36.41	36.41	32.88	32.88	9.41	36.41	32.88	34.65	34.65	O.K
226 D + 0.75XL1 + 0.525XEL(+Z)	76.02	0.99	0.00	33.78	1.76	0.00	35.55	35.55	32.02	32.02	8.55	35.55	32.02	33.79	33.79	O.K
227 D + 0.75XL + 0.525XEL(+X)	77.78	1.16	0.00	34.57	2.07	0.00	36.64	36.64	32.50	32.50	9.64	36.64	32.50	34.57	34.57	O.K
228 D + 0.75XL1 + 0.525XEL(+X)	75.85	1.16	0.00	33.71	2.07	0.00	35.78	35.78	31.64	31.64	8.78	35.78	31.64	33.71	33.71	O.K
229 D + 0.75XL + 0.525XEL(-Z)	77.61	0.99	0.00	34.49	1.76	0.00	36.26	36.26	32.73	32.73	9.26	36.26	32.73	34.50	34.50	O.K
230 D + 0.75XL1 + 0.525XEL(-Z)	77.95	0.99	0.00	34.64	1.76	0.00	36.41	36.41	32.88	32.88	9.41	36.41	32.88	34.65	34.65	O.K
231 D + 0.75XL + 0.525XEL(-X)	76.02	0.99	0.00	33.78	1.76	0.00	35.55	35.55	32.02	32.02	8.55	35.55	32.02	33.79	33.79	O.K
232 D + 0.75XL1 + 0.525XEL(-X)	77.78	1.16	0.00	34.57	2.07	0.00	36.64	36.64	32.50	32.50	9.64	36.64	32.50	34.57	34.57	O.K
233 0.6XD + WL(+Z)	75.85	1.16	0.00	33.71	2.07	0.00	35.78	35.78	31.64	31.64	8.78	35.78	31.64	33.71	33.71	O.K
234 0.6XD + WL(+X)	77.61	0.99	0.00	34.49	1.76	0.00	36.26	36.26	32.73	32.73	9.26	36.26	32.73	34.50	34.50	O.K
235 0.6XD + WL(-Z)	75.68	0.99	0.00	33.64	1.76	0.00	35.40	35.40	31.87	31.87	8.40	35.40	31.87	33.64	33.64	O.K
236 0.6XD + WL(-X)	77.78	0.82	0.00	34.57	1.46	0.00	36.03	36.03	33.11	33.11	9.03	36.03	33.11	34.57	34.57	O.K
237 0.6XD + 0.7XEL(+Z)	75.85	0.82	0.00	33.71	1.46	0.00	35.17	35.17	32.25	32.25	8.17	35.17	32.25	33.71	33.71	O.K
238 0.6XD + 0.7XEL(+X)	72.87	0.00	0.00	32.39	0.00	0.00	32.39	32.39	32.39	32.39	5.39	32.39	32.39	32.39	32.39	O.K
239 0.6XD + 0.7XEL(-Z)	69.92	3.03	0.00	31.07	5.38	0.00	36.45	36.45	25.70	25.70	9.45	36.45	25.70	31.08	31.08	O.K
240 0.6XD + 0.7XEL(-X)	66.97	0.00	0.00	29.76	0.00	0.00	29.76	29.76	29.76	29.76	2.76	29.76	29.76	29.76	29.76	O.K

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Strength Design

100 1.4XD	70.93	0.00	0.00	31.52	0.00	0.00	31.52	31.52	31.52	31.52	4.52	31.52	31.52	31.52	31.52
101 1.2XD + 1.2XTL1 + 1.6XL	86.37	2.11	0.00	38.39	3.76	0.00	42.15	42.15	34.63	34.63	15.15	42.15	34.63	38.39	38.39
102 1.2XD + 1.2XTL1 + 1.6XL1	82.25	2.11	0.00	36.55	3.76	0.00	40.31	40.31	32.80	32.80	13.31	40.31	32.80	36.56	36.56
103 1.2XD + 1.2XTL2 + 1.6XL	86.37	2.11	0.00	38.39	3.76	0.00	42.15	42.15	34.63	34.63	15.15	42.15	34.63	38.39	38.39
104 1.2XD + 1.2XTL2 + 1.6XL1	82.25	2.11	0.00	36.55	3.76	0.00	40.31	40.31	32.80	32.80	13.31	40.31	32.80	36.56	36.56
105 1.2XD + 0.8XWL(+Z)	73.04	0.00	0.00	32.46	0.00	0.00	32.46	32.46	32.46	32.46	5.46	32.46	32.46	32.46	32.46
106 1.2XD + 0.8XWL(+X)	70.68	2.42	0.00	31.41	4.30	0.00	35.71	35.71	27.11	27.11	8.71	35.71	27.11	31.41	31.41
107 1.2XD + 0.8XWL(-Z)	68.31	0.00	0.00	30.36	0.00	0.00	30.36	30.36	30.36	30.36	3.36	30.36	30.36	30.36	30.36
108 1.2XD + 0.8XWL(-X)	70.68	-2.42	0.00	31.41	4.30	0.00	35.71	35.71	27.11	27.11	8.71	35.71	27.11	31.41	31.41
109 0.9XD + 1.6XWL(+Z)	75.02	0.00	0.00	33.34	0.00	0.00	33.34	33.34	33.34	33.34	6.34	33.34	33.34	33.34	33.34
110 0.9XD + 1.6XWL(+X)	70.30	4.84	0.00	31.24	8.60	0.00	39.85	39.85	22.64	22.64	12.85	39.85	22.64	31.25	31.25
111 0.9XD + 1.6XWL(-Z)	65.57	0.00	0.00	29.14	0.00	0.00	29.14	29.14	29.14	29.14	2.14	29.14	29.14	29.14	29.14
112 0.9XD + 1.6XWL(-X)	70.30	-4.84	0.00	31.24	8.60	0.00	39.85	39.85	22.64	22.64	12.85	39.85	22.64	31.25	31.25
113 1.2XD + 0.5XL + 1.6XWL (+Z)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
114 1.2XD + 0.5XL + 1.6XWL (+X)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
115 1.2XD + 0.5XL + 1.6XWL (-Z)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
116 1.2XD + 0.5XL + 1.6XWL (-X)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
117 1.2XD + 0.5XL1 + 1.6XWL (+Z)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
118 1.2XD + 0.5XL1 + 1.6XWL (+X)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
121 1.2XD + 0.5XL + EL (+Z)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
122 1.2XD + 0.5XL + EL (+X)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
123 1.2XD + 0.5XL + EL (-Z)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
124 1.2XD + 0.5XL + EL (-X)	75.58	0.66	0.00	33.59	1.18	0.00	34.77	34.77	32.42	32.42	7.77	34.77	32.42	33.60	33.60
125 1.2XD + 0.5XL1 + EL (+Z)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
126 1.2XD + 0.5XL1 + EL (+X)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
127 1.2XD + 0.5XL1 + EL (-Z)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
128 1.2XD + 0.5XL1 + EL (-X)	74.29	0.66	0.00	33.02	1.18	0.00	34.19	34.19	31.84	31.84	7.19	34.19	31.84	33.02	33.02
129 0.9XD + EL(+Z)	70.62	0.00	0.00	31.39	0.00	0.00	31.39	31.39	31.39	31.39	4.39	31.39	31.39	31.39	31.39
130 0.9XD + EL(+X)	70.30	0.33	0.00	31.24	0.58	0.00	31.83	31.83	30.66	30.66	4.83	31.83	30.66	31.25	31.25
131 0.9XD + EL(-Z)	69.98	0.00	0.00	31.10	0.00	0.00	31.10	31.10	31.10	31.10	4.10	31.10	31.10	31.10	31.10
132 0.9XD + EL(-X)	70.30	-0.33	0.00	31.24	0.58	0.00	31.83	31.83	30.66	30.66	4.83	31.83	30.66	31.25	31.25
Max :	86	5	0												
Min :	66	-5	0												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 9.86 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 19.86 kN/m²
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 13.36 kN/m²
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 23.36 kN/m²

Sample calculation of values of base pressure:

Load case : 202 D + L1 + TL1

Maximum pressure, p_1 (pmax) = 34.51+2.35+0.00= 36.86 kN/m²
p_2= 34.51+2.35-0.00= 36.86 kN/m²
p_3= 34.51-(2.35+0.00)= 32.16 kN/m²
p_4= 34.51-2.35+0.00= 32.16 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 36.86-1.5*18= 9.86 kN/m²

Maximum average Pressure along X, $p_{avx[max]}$ = (36.86+36.86)/2= 36.86 kN/m²
Minimum average Pressure along X, $p_{avx[minx]}$ = (32.16+32.16)/2= 32.16 kN/m²
Maximum average Pressure along Y, $p_{avy[max]}$ = (36.86+32.16)/2= 34.51 kN/m²
Minimum average Pressure along Y, $p_{avy[minx]}$ = (32.16+36.86)/2= 34.51 kN/m²

Max PressureDL+LL	48.01	**OK**
Max Pressure Including SL/WL	50.36	**OK**

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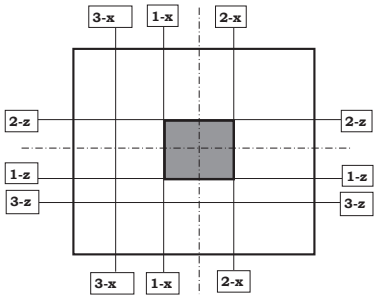
• DESIGN CALCULATION OF FOOTING MKD. GPS-6/ F1:

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.45+2*0.5*0.276)+(0.45+2*0.5*0.276))*2
= 2.904 m
Punching Area= ((0.45+2*0.5*0.276)*(0.45+2*0.5*0.276))
= 0.53 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)
= 29.10 kN/m^2
Area of footing, A_f= 2.25 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Conditions							
100 1.4XD	31.52	31.52	31.52	0.33	0.33	0.56	4.17
101 1.2XD + 1.2XTL1 + 1.6XL	39.52	37.26	40.98	1.68	0.88	2.90	15.98
102 1.2XD + 1.2XTL1 + 1.6XL1	37.68	35.43	39.14	1.42	0.63	2.48	12.82
103 1.2XD + 1.2XTL2 + 1.6XL	39.52	37.26	40.98	1.68	0.88	2.90	15.98
104 1.2XD + 1.2XTL2 + 1.6XL1	37.68	35.43	39.14	1.42	0.63	2.48	12.82
105 1.2XD + 0.8XWL(+Z)	32.46	32.46	32.46	0.46	0.46	0.78	5.78
106 1.2XD + 0.8XWL(+X)	32.70	30.12	34.37	0.77	-0.14	1.38	3.98
107 1.2XD + 0.8XWL(-Z)	30.36	30.36	30.36	0.17	0.17	0.29	2.17
108 1.2XD + 0.8XWL(-X)	32.70	30.12	34.37	0.77	-0.14	1.38	3.98
109 0.9XD + 1.6XWL(+Z)	33.34	33.34	33.34	0.58	0.58	0.99	7.30
110 0.9XD + 1.6XWL(+X)	33.83	28.66	37.18	1.20	-0.61	2.19	3.69
111 0.9XD + 1.6XWL(-Z)	29.14	29.14	29.14	0.01	0.01	0.01	0.08
112 0.9XD + 1.6XWL(-X)	33.83	28.66	37.18	1.20	-0.61	2.19	3.69
113 1.2XD + 0.5XL + 1.6XWL (+Z)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
114 1.2XD + 0.5XL + 1.6XWL (+X)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
115 1.2XD + 0.5XL + 1.6XWL (-Z)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
116 1.2XD + 0.5XL + 1.6XWL (-X)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
117 1.2XD + 0.5XL1 + 1.6XWL (+Z)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
118 1.2XD + 0.5XL1 + 1.6XWL (+X)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
121 1.2XD + 0.5XL + EL (+Z)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
122 1.2XD + 0.5XL + EL (+X)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
123 1.2XD + 0.5XL + EL (-Z)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
124 1.2XD + 0.5XL + EL (-X)	33.95	33.24	34.40	0.74	0.50	1.28	7.73
125 1.2XD + 0.5XL1 + EL (+Z)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
126 1.2XD + 0.5XL1 + EL (+X)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
127 1.2XD + 0.5XL1 + EL (-Z)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
128 1.2XD + 0.5XL1 + EL (-X)	33.37	32.66	33.82	0.66	0.42	1.14	6.74
129 0.9XD + EL(+Z)	31.39	31.39	31.39	0.32	0.32	0.53	3.93
131 0.9XD + EL(-Z)	31.10	31.10	31.10	0.28	0.28	0.47	3.44
132 0.9XD + EL(-X)	31.42	31.07	31.65	0.36	0.23	0.61	3.69

Sample calculation of BM & SF

Load case : 100 1.4XD

p_{1x} = 31.52-(31.52-31.52)*0.525/1.5 = 31.52 kN/m^2 Overhang = 0.525 m
p_{2x} = 31.52+(31.52-31.52)*0.525/1.5 = 31.52 kN/m^2 Effective depth = 0.292 m
p_{3x} = 31.52-(31.52-31.52)*(0.525-0.292)/1.5 = 31.52 kN/m^2
BM_{1x} = (31.52-29.1)*0.525*0.525/2+0.5*(31.52-31.52)*0.525*0.525*0.67 = 0.33 kN-m/m
BM_{2x} = (31.52-29.1)*0.525*0.525/2-0.5*(-31.52+31.52)*0.525*0.525*0.67 = 0.33 kN-m/m
SF_{3x} = (31.52-29.1)*(0.525-0.292)+0.5*(31.52-31.52)*(0.525-0.292) = 0.56 kN/m
S.$F_{(punching)}$ = (2.25-0.53)*(31.52-29.1) = 4.17 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.68	0.02	0.03	0.000	18.25	525.60	16ϕT	382.54	300	200	1005.31
At Top	-0.61	-0.01	0.01	0.000	6.67	525.60	16ϕT	382.54	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 2.90 kN/m
Shear stress , τ_v= (2.9*1000)/(1000*292)
= 0.01 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 15.98 kN
Strength of Concrete in shear Vc1 = 2508.32 kN **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 1676.82 kN **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 1672.22 kN **Table 22.6.5.2_ (c)**

Design Vc= 1672.22 kN
Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (15.98*1000)/(2904*292)
= 0.02 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M_{1z} kNm	B.M_{2z} kNm	S.F_{3z} kN
Ultimate Conditions						
100 1.4XD	31.52	31.52	31.52	0.33	0.33	0.60
101 1.2XD + 1.2XTL1 + 1.6XL	38.39	38.39	38.39	1.28	1.28	2.31
102 1.2XD + 1.2XTL1 + 1.6XL1	36.56	36.56	36.56	1.03	1.03	1.86
103 1.2XD + 1.2XTL2 + 1.6XL	38.39	38.39	38.39	1.28	1.28	2.31
104 1.2XD + 1.2XTL2 + 1.6XL1	36.56	36.56	36.56	1.03	1.03	1.86
105 1.2XD + 0.8XWL(+Z)	32.46	32.46	32.46	0.46	0.46	0.84
106 1.2XD + 0.8XWL(+X)	31.41	31.41	31.41	0.32	0.32	0.58
107 1.2XD + 0.8XWL(-Z)	30.36	30.36	30.36	0.17	0.17	0.31
108 1.2XD + 0.8XWL(-X)	31.41	31.41	31.41	0.32	0.32	0.58
109 0.9XD + 1.6XWL(+Z)	33.34	33.34	33.34	0.58	0.58	1.06
110 0.9XD + 1.6XWL(+X)	31.25	31.25	31.25	0.30	0.30	0.54
111 0.9XD + 1.6XWL(-Z)	29.14	29.14	29.14	0.01	0.01	0.01
112 0.9XD + 1.6XWL(-X)	31.25	31.25	31.25	0.30	0.30	0.54
113 1.2XD + 0.5XL + 1.6XWL (+Z)	33.60	33.60	33.60	0.62	0.62	1.12
114 1.2XD + 0.5XL + 1.6XWL (+X)	33.60	33.60	33.60	0.62	0.62	1.12
115 1.2XD + 0.5XL + 1.6XWL (-Z)	33.60	33.60	33.60	0.62	0.62	1.12
116 1.2XD + 0.5XL + 1.6XWL (-X)	33.60	33.60	33.60	0.62	0.62	1.12
117 1.2XD + 0.5XL1 + 1.6XWL (+Z)	33.02	33.02	33.02	0.54	0.54	0.98
118 1.2XD + 0.5XL1 + 1.6XWL (+X)	33.02	33.02	33.02	0.54	0.54	0.98
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	33.02	33.02	33.02	0.54	0.54	0.98
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	33.02	33.02	33.02	0.54	0.54	0.98
121 1.2XD + 0.5XL + EL (+Z)	33.60	33.60	33.60	0.62	0.62	1.12
122 1.2XD + 0.5XL + EL (+X)	33.60	33.60	33.60	0.62	0.62	1.12
123 1.2XD + 0.5XL + EL (-Z)	33.60	33.60	33.60	0.62	0.62	1.12
124 1.2XD + 0.5XL + EL (-X)	33.60	33.60	33.60	0.62	0.62	1.12
125 1.2XD + 0.5XL1 + EL (+Z)	33.02	33.02	33.02	0.54	0.54	0.98
126 1.2XD + 0.5XL1 + EL (+X)	33.02	33.02	33.02	0.54	0.54	0.98
127 1.2XD + 0.5XL1 + EL (-Z)	33.02	33.02	33.02	0.54	0.54	0.98
128 1.2XD + 0.5XL1 + EL (-X)	33.02	33.02	33.02	0.54	0.54	0.98
129 0.9XD + EL(+Z)	31.39	31.39	31.39	0.32	0.32	0.57
131 0.9XD + EL(-Z)	31.10	31.10	31.10	0.28	0.28	0.50
132 0.9XD + EL(-X)	31.25	31.25	31.25	0.30	0.30	0.54

Sample calculation of BM & SF

Load case : 100 1.4XD

p_{1z} = 31.52-(31.52-31.52)*0.525/1.5= 31.52 kN/m^2

p_{2z} = 31.52+(31.52-31.52)*0.525/1.5= 31.52 kN/m^2 Overhang = 0.525 m

p_{3z} = 31.52-(31.52-31.52)*(0.525-0.276)/1.5= 31.52 kN/m^2 Effective depth = 0.276 m

BM_{1z} = (31.52-29.1)*0.525*0.525/2+0.5*(31.52-31.52)*0.525*0.525*0.67 = 0.33 kN-m/m

BM_{2z} = (31.52-29.1)*0.525*0.525/2-0.5*(-31.52+31.52)*0.525*0.525*0.67 = 0.33 kN-m/m

SF_{3z} = (31.52-29.1)*(0.525-0.276)+0.5*(31.52-31.52)*(0.525-0.276) = 0.60 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.28	0.017	0.022	0.0001	15.59	496.80	16ϕT	404.71	300	150	1340.4
Top	0.00	0.000	0.000	0.0000	0.00	496.80	16ϕT	404.71	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 2.31 kN/m

Shear stress , τ_v= (2.31*1000)/(1000*276)

= 0.01 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c

At Top **16ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **150** c/c

At Top **16ϕT** **200** c/c

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- **DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F2:**

Size of Columns at node	Along-X	Along-Z
Col.1	0.50m	0.50m
Col.2	0.50m	0.50m
Size of fdn. raft:	3.00m	1.50m
Thickness of raft:	0.35 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.50 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation:

Weight of raft = (3*1.5*0.35*25) = 39.38KN

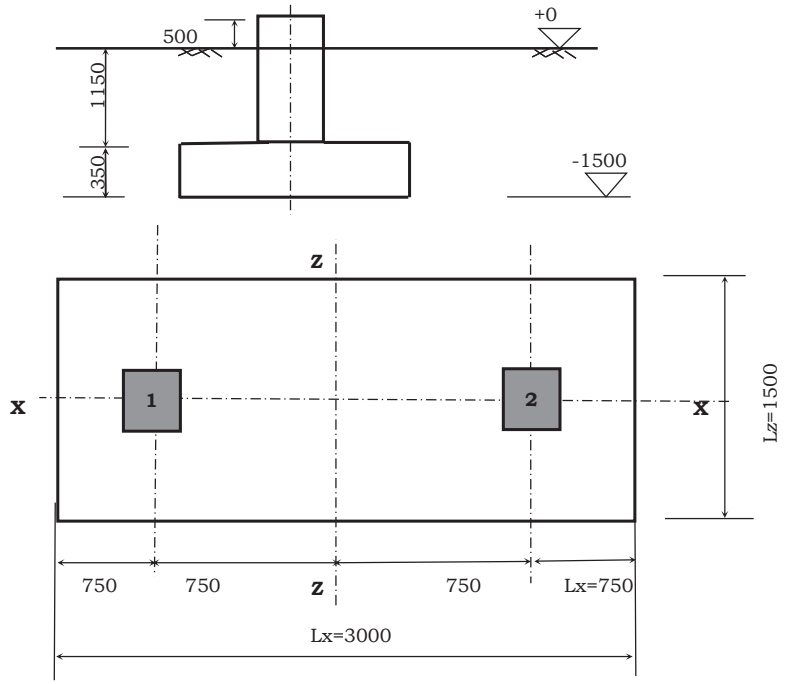
Weight of soil = ((3*1.5)-2*(0.5*0.5))*1.15*20 = 92.00KN

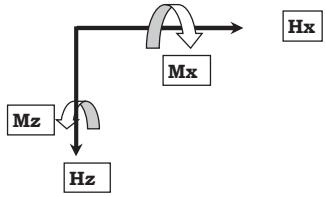
Total = 131.38KN

Submerged wt. = 131.38-3*1.5*1.5*10 = 63.88KN

Design Input:

Characteristic strength of concrete, f'_c =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm





Hx=Force along X-X direction

Hz=Force along Z-Z direction

Mx=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F2:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	46.24	0
Min Px	23	201	0	46.24	0
Max Py	235	202	0	77.92	0
Min Py	173	236	0	14.96	0
Max Pz	23	201	0	46.24	0
Min Pz	23	201	0	46.24	0

Gross Maximum Base pressure, p_{max} = 77.9 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 480 KN/m^2
> 77.9 KN/m^2
Hence OK

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F2:

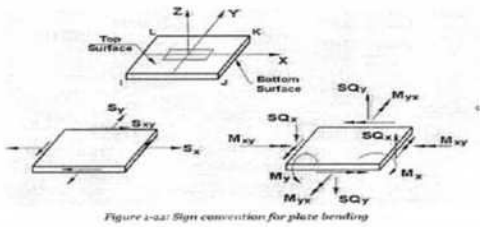
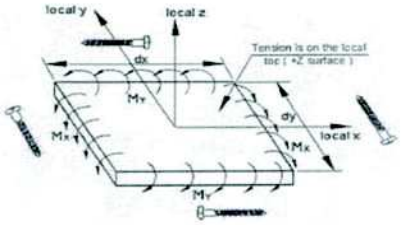


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	121	0.203	-0.027	0	0	0	-17.62	-2.998	-2.551	-20.173	-5.549	103 1.2XD + 1.2XTL2 + 1.6XL
Min Qx	69	-0.203	-0.027	0	0	0	-17.62	-2.998	0	-17.622	-2.998	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Qy	110	0.003	0.123	0	0	0	-2.286	-3.797	-0.36	-2.646	-4.157	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	113	0.049	-0.15	0	0	0	-3.397	-6.5	1.505	-4.902	-8.005	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Sx	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sx	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Sy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Sxy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sxy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Mx	179	0.087	-0.03	0	0	0	35.027	2.73	-1.695	36.722	4.425	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Mx	180	-0.165	-0.043	0	0	0	-38.14	-10.587	-0.615	-38.752	-11.202	103 1.2XD + 1.2XTL2 + 1.6XL
Max My	101	-0.021	-0.024	0	0	0	0.581	4.869	0.676	1.257	5.545	103 1.2XD + 1.2XTL2 + 1.6XL
Min My	111	0.136	0.087	0	0	0	-7.949	-11.744	0.448	-8.397	-12.192	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Mxy	110	-0.009	-0.112	0	0	0	0.312	3.382	3.562	3.874	6.944	103 1.2XD + 1.2XTL2 + 1.6XL
Min Mxy	58	0.009	-0.112	0	0	0	0.312	3.382	-3.562	3.874	6.944	104 1.2XD + 1.2XTL2 + 1.6XL1

PLATE CENTER STRESS : FDN_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	134	0.079	-0.007	0	0	0	-13.24	-0.659	-2.095	-15.337	-2.754	103 1.2XD + 1.2XTL2 + 1.6XL
Min Qx	91	-0.079	-0.007	0	0	0	-13.24	-0.659	0	-13.242	-0.659	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Qy	104	0.011	0.043	0	0	0	-0.693	-4.705	0.807	-1.5	-5.512	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	102	0.017	-0.054	0	0	0	-0.673	-5.676	-0.274	-0.947	-5.95	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Sx	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sx	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Sy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Sxy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Min Sxy	206	-0.001	0.002	0	0	0	-0.204	-0.19	-0.091	-0.295	-0.281	100 1.4XD
Max Mx	188	0.035	0.014	0	0	0	29.741	2.673	-0.393	30.134	3.066	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Mx	186	-0.029	0.043	0	0	0	-29.94	-4.037	-0.673	-30.616	-4.71	103 1.2XD + 1.2XTL2 + 1.6XL
Max My	101	-0.021	-0.024	0	0	0	0.581	4.869	0.676	1.257	5.545	103 1.2XD + 1.2XTL2 + 1.6XL
Min My	100	0.04	-0.032	0	0	0	-0.954	-10.257	0.272	-1.226	-10.529	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Mxy	198	0.071	-0.016	0	0	0	-11.16	0.194	2.146	-13.301	2.34	103 1.2XD + 1.2XTL2 + 1.6XL
Min Mxy	197	-0.071	-0.016	0	0	0	-11.16	0.194	-2.146	-13.301	2.34	104 1.2XD + 1.2XTL2 + 1.6XL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		350	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

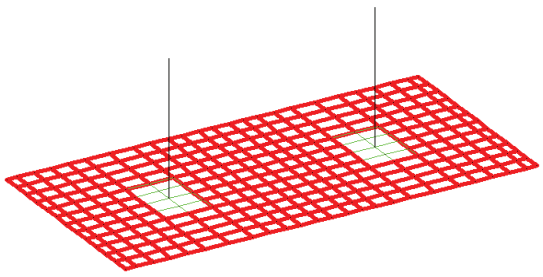
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102 1.2X	36.72	top face	50	292	36.722	0.57	0.0014	403.2	525.6	16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	-38.75	bottom face	50	292	38.752	0.61	0.0015	425.7		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	3.87	top face	50	292	3.874	0.06	0.0001	42.2		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
104 1.2X	3.87	top face	50	292	3.874	0.06	0.0001	42.2		16	382.69	200	1005.8	14.18	17.73	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.079	1	27.65	15.337	0.003	0.292	285.4	293.498	500.974	285.4	1069	> 27.65	ok	214.1	φVc>Vu	not required
	Min Qx	0.079	1	27.65	13.242			286.824			286.82	1070	> 27.65	ok	215.1	φVc>Vu	not required

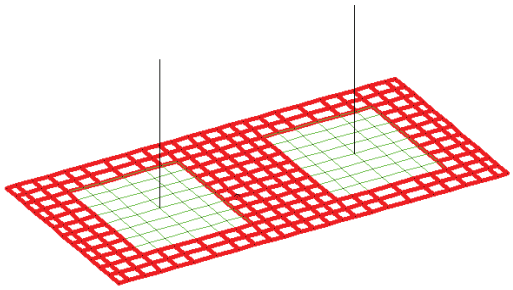
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2X	5.55	top face	50	292	5.545	0.09	0.0002	60.4	525.6	16	382.69	200	1005.8	14.18	17.73	0.046	tension control
102 1.2X	-12.19	bottom face	50	292	12.192	0.19	0.0005	133.0		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	6.94	top face	50	292	6.944	0.11	0.0003	75.6		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
104 1.2X	6.94	top face	50	292	6.944	0.11	0.0003	75.6		16	382.69	200	1005.8	14.18	17.73	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	102 1.2X	0.043	1	15.05	5.512	0.003	0.292	290.032	293.498	fc'501	290.03	303	> 15.05	ok	217.5	φVc>Vu	not required
	102 1.2X	0.054	1	18.9	5.95			292.259			292.26	305	> 18.9	ok	219.2	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F2:

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
100 1.4XD	3.4	0.0	0.0	0.0	0.0	3.4	0.0	0.0	0.0	0.0
101 1.2XD + 1.2XTL1 + 1.6XL	4.2	-6.4	0.0	-30.4	0.0	3.3	-6.4	0.0	16.0	0.0
102 1.2XD + 1.2XTL1 + 1.6XL1	3.3	-6.4	0.0	-16.0	0.0	4.2	-6.4	0.0	30.4	0.0
103 1.2XD + 1.2XTL2 + 1.6XL	4.3	-6.4	0.0	16.4	0.0	3.4	-6.4	0.0	-30.8	0.0
104 1.2XD + 1.2XTL2 + 1.6XL1	3.4	-6.4	0.0	30.8	0.0	4.3	-6.4	0.0	-16.4	0.0
105 1.2XD + 0.8XWL(+Z)	3.0	0.0	0.0	-1.1	0.0	2.9	0.0	0.0	-1.1	0.0
106 1.2XD + 0.8XWL(+X)	3.0	-1.1	0.0	0.0	0.0	3.0	-1.1	0.0	0.0	0.0
107 1.2XD + 0.8XWL(-Z)	2.9	0.0	0.0	1.1	0.0	3.0	0.0	0.0	1.1	0.0
108 1.2XD + 0.8XWL(-X)	3.0	1.1	0.0	0.0	0.0	3.0	1.1	0.0	0.0	0.0
109 0.9XD + 1.6XWL(+Z)	2.4	0.0	0.0	-2.2	0.0	2.1	0.0	0.0	-2.2	0.0
110 0.9XD + 1.6XWL(+X)	2.2	-2.2	0.0	0.0	0.0	2.2	-2.2	0.0	0.0	0.0
111 0.9XD + 1.6XWL(-Z)	2.1	0.0	0.0	2.2	0.0	2.4	0.0	0.0	2.2	0.0
112 0.9XD + 1.6XWL(-X)	2.2	2.2	0.0	0.0	0.0	2.2	2.2	0.0	0.0	0.0
113 1.2XD + 0.5XL + 1.6XWL (+Z)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
114 1.2XD + 0.5XL + 1.6XWL (+X)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
115 1.2XD + 0.5XL + 1.6XWL (-Z)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
116 1.2XD + 0.5XL + 1.6XWL (-X)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
117 1.2XD + 0.5XL1 + 1.6XWL (+Z	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
118 1.2XD + 0.5XL1 + 1.6XWL (+X	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
121 1.2XD + 0.5XL + EL (+Z)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
122 1.2XD + 0.5XL + EL (+X)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
123 1.2XD + 0.5XL + EL (-Z)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
124 1.2XD + 0.5XL + EL (-X)	3.3	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
125 1.2XD + 0.5XL1 + EL (+Z)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
126 1.2XD + 0.5XL1 + EL (+X)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
127 1.2XD + 0.5XL1 + EL (-Z)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
128 1.2XD + 0.5XL1 + EL (-X)	3.1	-2.0	0.0	2.3	0.0	3.3	-2.0	0.0	2.2	0.0
129 0.9XD + EL(+Z)	2.2	0.0	0.0	-0.1	0.0	2.2	0.0	0.0	-0.1	0.0
130 0.9XD + EL(+X)	2.2	-0.1	0.0	0.0	0.0	2.2	-0.1	0.0	0.0	0.0
131 0.9XD + EL(-Z)	2.2	0.0	0.0	0.1	0.0	2.2	0.0	0.0	0.1	0.0
132 0.9XD + EL(-X)	2.2	0.1	0.0	0.0	0.0	2.2	0.1	0.0	0.0	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
100 1.4XD	138.3	0.00	0.0	0.0	0.0	NA	NA	NA	NA
101 1.2XD + 1.2XTL1 + 1.6XL	138.8	-12.80	-0.7	-14.4	0.0	311.3	NA	4.3	3.9
102 1.2XD + 1.2XTL1 + 1.6XL1	138.8	-12.80	0.7	14.4	0.0	311.3	NA	4.3	3.9
103 1.2XD + 1.2XTL2 + 1.6XL	139.0	-12.80	-0.7	-14.4	0.0	311.3	NA	4.3	3.9
104 1.2XD + 1.2XTL2 + 1.6XL1	139.0	-12.80	0.7	14.4	0.0	311.3	NA	4.3	3.9
105 1.2XD + 0.8XWL(+Z)	137.3	0.00	-0.1	-2.2	0.0	1920.2	NA	NA	25.0
106 1.2XD + 0.8XWL(+X)	137.3	-2.20	0.0	0.0	0.0	NA	NA	25.0	NA
107 1.2XD + 0.8XWL(-Z)	137.3	0.00	0.1	2.2	0.0	1920.2	NA	NA	25.0
108 1.2XD + 0.8XWL(-X)	137.3	2.20	0.0	0.0	0.0	NA	NA	25.0	NA
109 0.9XD + 1.6XWL(+Z)	135.8	0.00	-0.2	-4.4	0.0	949.7	NA	NA	12.3
110 0.9XD + 1.6XWL(+X)	135.8	-4.40	0.0	0.0	0.0	NA	NA	12.3	NA
111 0.9XD + 1.6XWL(-Z)	135.8	0.00	0.2	4.4	0.0	949.7	NA	NA	12.3
112 0.9XD + 1.6XWL(-X)	135.8	4.40	0.0	0.0	0.0	NA	NA	12.3	NA
113 1.2XD + 0.5XL + 1.6XWL (+Z)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
114 1.2XD + 0.5XL + 1.6XWL (+X)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
115 1.2XD + 0.5XL + 1.6XWL (-Z)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
116 1.2XD + 0.5XL + 1.6XWL (-X)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
117 1.2XD + 0.5XL1 + 1.6XWL (+Z	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
118 1.2XD + 0.5XL1 + 1.6XWL (+X	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
121 1.2XD + 0.5XL + EL (+Z)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
122 1.2XD + 0.5XL + EL (+X)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
123 1.2XD + 0.5XL + EL (-Z)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
124 1.2XD + 0.5XL + EL (-X)	137.8	-4.00	-0.2	-4.5	0.0	991.4	NA	13.8	12.2
125 1.2XD + 0.5XL1 + EL (+Z)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
126 1.2XD + 0.5XL1 + EL (+X)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
127 1.2XD + 0.5XL1 + EL (-Z)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
128 1.2XD + 0.5XL1 + EL (-X)	137.8	-4.00	0.2	4.5	0.0	991.4	NA	13.8	12.2
129 0.9XD + EL(+Z)	135.8	0.00	0.0	-0.3	0.0	15090.4	NA	NA	195.4
130 0.9XD + EL(+X)	135.8	-0.28	0.0	0.0	0.0	NA	NA	195.4	NA
131 0.9XD + EL(-Z)	135.8	0.00	0.0	0.3	0.0	15090.4	NA	NA	195.4
132 0.9XD + EL(-X)	135.8	0.28	0.0	0.0	0.0	NA	NA	195.4	NA
					Min.=	311.3	0.0	4.3	3.9

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- **DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F7:**

Size of Columns at node	Along-X	Along-Z
Col.1	0.50m	0.50m
Col.2	0.50m	0.50m
Size of fdn. raft:	2.85m	1.50m
Thickness of raft:	0.35 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.50 m	
U/S level of fdn:	-1.50 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation:

Weight of raft = (2.85*1.5*0.35*25) = 37.41KN

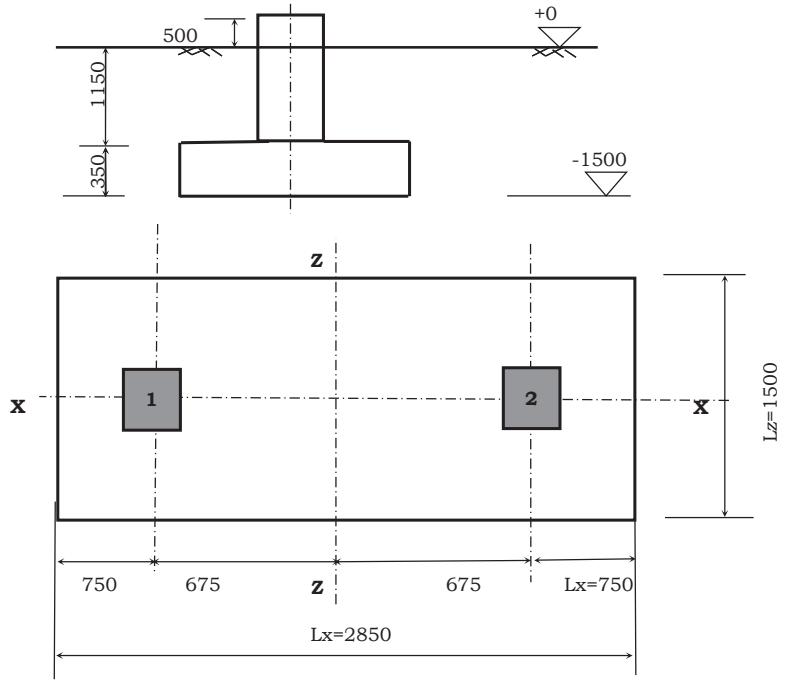
Weight of soil = ((2.85*1.5)-2*(0.5*0.5))*1.15*20 = 86.83KN

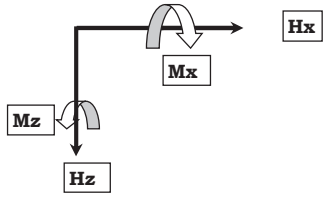
Total = 124.23KN

Submerged wt. = 124.23-2.85*1.5*1.5*10 = 60.11KN

Design Input:

Characteristic strength of concrete, f'_c =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm





Hx=Force along X-X direction

Hz=Force along Z-Z direction

Mx=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F7:

Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	23	201	0	44.46	0
Min Px	23	201	0	44.46	0
Max Py	173	201	0	77.43	0
Min Py	308	203	0	16.03	0
Max Pz	23	201	0	44.46	0
Min Pz	23	201	0	44.46	0

Gross Maximum Base pressure, p_{max} = 77.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 1.5 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 480 KN/m^2
> 77.4 KN/m^2
Hence OK

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F7:

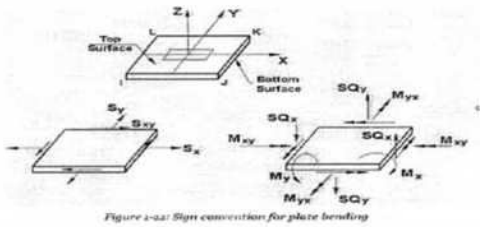
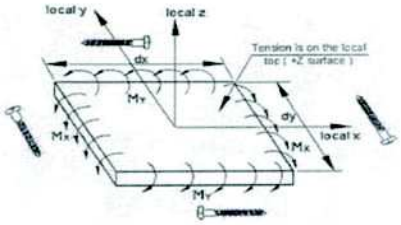


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	121	0.205	-0.028	0	0	0	-17.76	-3.133	-2.538	-20.298	-5.671	103 1.2XD + 1.2XTL2 + 1.6XL
Min Qx	69	-0.205	-0.028	0	0	0	-17.76	-3.133	0	-17.76	-3.133	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Qy	110	0.004	0.123	0	0	0	-2.395	-3.772	-0.305	-2.7	-4.077	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Qy	181	-0.055	-0.151	0	0	0	-33.67	-6.523	-1.489	-35.159	-8.012	103 1.2XD + 1.2XTL2 + 1.6XL
Max Sx	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sx	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Sy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Sxy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sxy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Mx	179	0.089	-0.03	0	0	0	34.382	2.86	-1.561	35.943	4.421	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Mx	180	-0.158	-0.044	0	0	0	-38.69	-11.235	-0.575	-39.265	-11.81	103 1.2XD + 1.2XTL2 + 1.6XL
Max My	101	-0.021	-0.025	0	0	0	0.572	4.758	0.736	1.308	5.494	103 1.2XD + 1.2XTL2 + 1.6XL
Min My	111	0.137	0.087	0	0	0	-8.043	-11.691	0.529	-8.572	-12.22	102 1.2XD + 1.2XTL1 + 1.6XL1
Max Mxy	110	-0.008	-0.112	0	0	0	0.213	3.28	3.685	3.898	6.965	103 1.2XD + 1.2XTL2 + 1.6XL
Min Mxy	58	0.008	-0.112	0	0	0	0.213	3.28	-3.685	3.898	6.965	104 1.2XD + 1.2XTL2 + 1.6XL1

PLATE CENTER STRESS : FDN_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	134	0.081	-0.007	0	0	0	-13.39	-0.7	-2.098	-15.49	-2.798	103 1.2XD + 1.2XTL2 + 1.6XL
Min Qx	91	-0.081	-0.007	0	0	0	-13.39	-0.7	0	-13.392	-0.7	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Qy	162	0.027	0.053	0	0	0	-31.1	-4.337	0.441	-31.538	-4.778	104 1.2XD + 1.2XTL2 + 1.6XL1
Min Qy	165	0.035	-0.055	0	0	0	-30.13	-5.693	-0.523	-30.648	-6.216	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Sx	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sx	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Sy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Sxy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Min Sxy	206	-0.001	0.003	0	0	0	-0.641	-0.207	-0.068	-0.709	-0.275	100 1.4XD
Max Mx	188	0.039	0.019	0	0	0	29.435	2.757	-0.33	29.765	3.087	102 1.2XD + 1.2XTL1 + 1.6XL1
Min Mx	187	-0.057	0.032	0	0	0	-31.32	-9.237	-0.543	-31.864	-9.78	103 1.2XD + 1.2XTL2 + 1.6XL
Max My	101	-0.021	-0.025	0	0	0	0.572	4.758	0.736	1.308	5.494	103 1.2XD + 1.2XTL2 + 1.6XL
Min My	164	0.063	-0.029	0	0	0	-30.93	-10.432	-0.118	-31.044	-10.55	104 1.2XD + 1.2XTL2 + 1.6XL1
Max Mxy	198	0.07	-0.016	0	0	0	-11.42	0.159	2.27	-13.687	2.429	103 1.2XD + 1.2XTL2 + 1.6XL
Min Mxy	197	-0.07	-0.016	0	0	0	-11.42	0.159	-2.27	-13.687	2.429	104 1.2XD + 1.2XTL2 + 1.6XL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		350	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

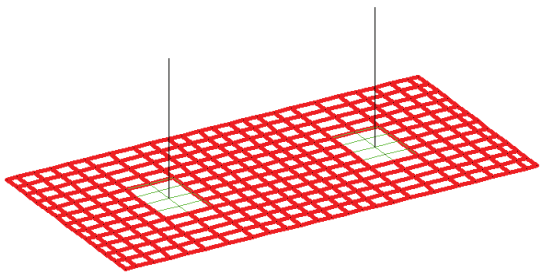
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
102 1.2X	35.94	top face	50	292	35.943	0.56	0.0014	394.5	525.6	16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	-39.27	bottom face	50	292	39.265	0.61	0.0015	431.4		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	3.90	top face	50	292	3.898	0.06	0.0001	42.4		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
104 1.2X	3.90	top face	50	292	3.898	0.06	0.0001	42.4		16	382.69	200	1005.8	14.18	17.73	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.081	1	28.35	15.49	0.003	0.292	285.537	293.498	500.974	285.54	1069	> 28.35	ok	214.2	φVc>Vu	not required
	Min Qx	0.081	1	28.35	13.392			286.969			286.97	1070	> 28.35	ok	215.2	φVc>Vu	not required

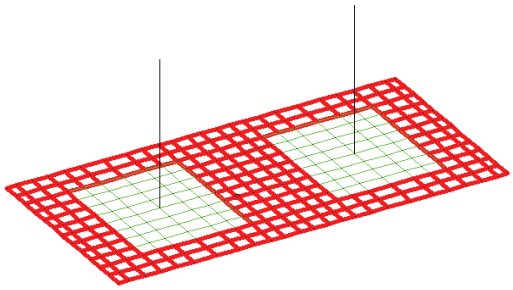
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2X	5.49	top face	50	292	5.494	0.09	0.0002	59.8	525.6	16	382.69	200	1005.8	14.18	17.73	0.046	tension control
102 1.2X	-12.22	bottom face	50	292	12.22	0.19	0.0005	133.3		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
103 1.2X	6.97	top face	50	292	6.965	0.11	0.0003	75.9		16	382.69	200	1005.8	14.18	17.73	0.046	tension control
104 1.2X	6.97	top face	50	292	6.965	0.11	0.0003	75.9		16	382.69	200	1005.8	14.18	17.73	0.046	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	φ(V_c +0.66√ f_c'bd)≥V_u		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	104 1.2X	0.053	1	18.55	4.778	0.003	0.292	295.783	293.498	fc'501	293.5	306	> 18.55	ok	220.1	φVc>Vu	not required
	104 1.2X	0.055	1	19.25	6.216			291.861			291.86	304	> 19.25	ok	218.9	φVc>Vu	not required



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

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DESIGN CALCULATION OF FOOTING MKD. GPS-7/ F7:

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
100 1.4XD	3.5	0.0	0.0	0.0	0.0	3.5	0.0	0.0	0.0	0.0
101 1.2XD + 1.2XTL1 + 1.6XL	4.2	-6.4	0.0	-30.4	0.0	3.3	-6.4	0.0	16.0	0.0
102 1.2XD + 1.2XTL1 + 1.6XL1	3.3	-6.4	0.0	-16.0	0.0	4.2	-6.4	0.0	30.4	0.0
103 1.2XD + 1.2XTL2 + 1.6XL	4.4	-6.4	0.0	16.4	0.0	3.4	-6.4	0.0	-30.8	0.0
104 1.2XD + 1.2XTL2 + 1.6XL1	3.4	-6.4	0.0	30.8	0.0	4.4	-6.4	0.0	-16.4	0.0
105 1.2XD + 0.8XWL(+Z)	3.1	0.0	0.0	-1.1	0.0	2.9	0.0	0.0	-1.1	0.0
106 1.2XD + 0.8XWL(+X)	3.0	-1.1	0.0	0.0	0.0	3.0	-1.1	0.0	0.0	0.0
107 1.2XD + 0.8XWL(-Z)	2.9	0.0	0.0	1.1	0.0	3.1	0.0	0.0	1.1	0.0
108 1.2XD + 0.8XWL(-X)	3.0	1.1	0.0	0.0	0.0	3.0	1.1	0.0	0.0	0.0
109 0.9XD + 1.6XWL(+Z)	2.4	0.0	0.0	-2.2	0.0	2.1	0.0	0.0	-2.2	0.0
110 0.9XD + 1.6XWL(+X)	2.2	-2.2	0.0	0.0	0.0	2.2	-2.2	0.0	0.0	0.0
111 0.9XD + 1.6XWL(-Z)	2.1	0.0	0.0	2.2	0.0	2.4	0.0	0.0	2.2	0.0
112 0.9XD + 1.6XWL(-X)	2.2	2.2	0.0	0.0	0.0	2.2	2.2	0.0	0.0	0.0
113 1.2XD + 0.5XL + 1.6XWL (+Z)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
114 1.2XD + 0.5XL + 1.6XWL (+X)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
115 1.2XD + 0.5XL + 1.6XWL (-Z)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
116 1.2XD + 0.5XL + 1.6XWL (-X)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
117 1.2XD + 0.5XL1 + 1.6XWL (+Z	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
118 1.2XD + 0.5XL1 + 1.6XWL (+X	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
121 1.2XD + 0.5XL + EL (+Z)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
122 1.2XD + 0.5XL + EL (+X)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
123 1.2XD + 0.5XL + EL (-Z)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
124 1.2XD + 0.5XL + EL (-X)	3.4	-2.0	0.0	-2.2	0.0	3.1	-2.0	0.0	-2.3	0.0
125 1.2XD + 0.5XL1 + EL (+Z)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
126 1.2XD + 0.5XL1 + EL (+X)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
127 1.2XD + 0.5XL1 + EL (-Z)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
128 1.2XD + 0.5XL1 + EL (-X)	3.1	-2.0	0.0	2.3	0.0	3.4	-2.0	0.0	2.2	0.0
129 0.9XD + EL(+Z)	2.2	0.0	0.0	-0.1	0.0	2.2	0.0	0.0	-0.1	0.0
130 0.9XD + EL(+X)	2.2	-0.1	0.0	0.0	0.0	2.2	-0.1	0.0	0.0	0.0
131 0.9XD + EL(-Z)	2.2	0.0	0.0	0.1	0.0	2.2	0.0	0.0	0.1	0.0
132 0.9XD + EL(-X)	2.2	0.1	0.0	0.0	0.0	2.2	0.1	0.0	0.0	0.0

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
100 1.4XD	131.2	0.00	0.0	0.0	0.0	NA	NA	NA	NA
101 1.2XD + 1.2XTL1 + 1.6XL	131.8	-12.80	-0.6	-14.4	0.0	299.4	NA	4.1	3.7
102 1.2XD + 1.2XTL1 + 1.6XL1	131.8	-12.80	0.6	14.4	0.0	299.4	NA	4.1	3.7
103 1.2XD + 1.2XTL2 + 1.6XL	132.0	-12.80	-0.6	-14.4	0.0	299.7	NA	4.1	3.7
104 1.2XD + 1.2XTL2 + 1.6XL1	132.0	-12.80	0.6	14.4	0.0	299.7	NA	4.1	3.7
105 1.2XD + 0.8XWL(+Z)	130.2	0.00	-0.1	-2.2	0.0	1857.2	NA	NA	23.7
106 1.2XD + 0.8XWL(+X)	130.2	-2.20	0.0	0.0	0.0	NA	NA	23.7	NA
107 1.2XD + 0.8XWL(-Z)	130.2	0.00	0.1	2.2	0.0	1857.2	NA	NA	23.7
108 1.2XD + 0.8XWL(-X)	130.2	2.20	0.0	0.0	0.0	NA	NA	23.7	NA
109 0.9XD + 1.6XWL(+Z)	128.7	0.00	-0.2	-4.4	0.0	917.9	NA	NA	11.7
110 0.9XD + 1.6XWL(+X)	128.7	-4.40	0.0	0.0	0.0	NA	NA	11.7	NA
111 0.9XD + 1.6XWL(-Z)	128.7	0.00	0.2	4.4	0.0	917.9	NA	NA	11.7
112 0.9XD + 1.6XWL(-X)	128.7	4.40	0.0	0.0	0.0	NA	NA	11.7	NA
113 1.2XD + 0.5XL + 1.6XWL (+Z)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
114 1.2XD + 0.5XL + 1.6XWL (+X)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
115 1.2XD + 0.5XL + 1.6XWL (-Z)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
116 1.2XD + 0.5XL + 1.6XWL (-X)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
117 1.2XD + 0.5XL1 + 1.6XWL (+Z	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
118 1.2XD + 0.5XL1 + 1.6XWL (+X	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
119 1.2XD + 0.5XL1 + 1.6XWL (-Z)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
120 1.2XD + 0.5XL1 + 1.6XWL (-X)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
121 1.2XD + 0.5XL + EL (+Z)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
122 1.2XD + 0.5XL + EL (+X)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
123 1.2XD + 0.5XL + EL (-Z)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
124 1.2XD + 0.5XL + EL (-X)	130.7	-4.00	-0.2	-4.5	0.0	948.4	NA	13.1	11.6
125 1.2XD + 0.5XL1 + EL (+Z)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
126 1.2XD + 0.5XL1 + EL (+X)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
127 1.2XD + 0.5XL1 + EL (-Z)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
128 1.2XD + 0.5XL1 + EL (-X)	130.7	-4.00	0.2	4.5	0.0	948.4	NA	13.1	11.6
129 0.9XD + EL(+Z)	128.7	0.00	0.0	-0.3	0.0	15094.9	NA	NA	185.2
130 0.9XD + EL(+X)	128.7	-0.28	0.0	0.0	0.0	NA	NA	185.2	NA
131 0.9XD + EL(-Z)	128.7	0.00	0.0	0.3	0.0	15094.9	NA	NA	185.2
132 0.9XD + EL(-X)	128.7	0.28	0.0	0.0	0.0	NA	NA	185.2	NA
					Min.=	299.4	0.0	4.1	3.7

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REFERENCES

[1] Saudi concrete structures Code; SBC 304-2018
[2] Building Code Requirements for Structural Concrete; ACI 318-14
[3] Saudi Building Code Requirements : Structural - Loading and Forces; SBC 301-2007

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ATTACHMENT 1 STAAD OUTPUT

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    OCT 19, 2020                 *
*           Time=    15:32:44                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\FOUNDATION\GPS-6 _7\WORKING\GPS-7 F2.STD
     2. START JOB INFORMATION
     3. *FOUNDATION F5
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 23 -0.25 -2 -3; 121 -0.25 -2 0; 122 -0.25 -2 -0.25; 123 0 -2 -0.25; 124 0 -2 0
    10. 125 -0.5 -2 0; 126 -0.5 -2 -0.25; 130 0.25 -2 -0.25; 131 0.25 -2 0
    11. 133 -0.75 -2 0; 134 -0.75 -2 -0.25; 136 -0.25 -2 -0.5; 137 0 -2 -0.5
    12. 138 -0.5 -2 -0.5; 139 0.25 -2 -0.5; 140 -0.75 -2 -0.5; 146 -0.75 -2 -0.75
    13. 147 -0.5 -2 -0.75; 148 -0.25 -2 -0.75; 149 0 -2 -0.75; 150 0.25 -2 -0.75
    14. 173 0.5 -2 0; 174 0.5 -2 -0.25; 175 0.5 -2 -0.5; 176 0.5 -2 -0.75; 186 0 -2 -3
    15. 187 -0.5 -2 -3; 189 -0.25 -2 -2.75; 190 0 -2 -2.75; 191 -0.5 -2 -2.75
    16. 192 0.25 -2 -3; 193 0.25 -2 -2.75; 195 -0.75 -2 -2.75; 196 -0.75 -2 -3
    17. 203 -0.75 -2 -2.5; 204 -0.5 -2 -2.5; 205 -0.25 -2 -2.5; 206 0 -2 -2.5
    18. 207 0.25 -2 -2.5; 213 -0.75 -2 -2.25; 214 -0.5 -2 -2.25; 215 -0.25 -2 -2.25
    19. 216 0 -2 -2.25; 217 0.25 -2 -2.25; 232 0.5 -2 -2.5; 233 0.5 -2 -2.25
    20. 234 0.5 -2 -2.75; 235 0.5 -2 -3; 246 -0.75 -2 -1; 247 -0.5 -2 -1.
    21. 248 -0.25 -2 -1; 249 0 -2 -1; 250 0.25 -2 -1; 253 0.5 -2 -1.
    22. 255 -0.75 -2 -1.25; 256 -0.5 -2 -1.25; 257 -0.25 -2 -1.25; 258 0 -2 -1.25
    23. 259 0.25 -2 -1.25; 262 0.5 -2 -1.25; 273 -0.75 -2 -2; 274 -0.5 -2 -2
    24. 275 -0.25 -2 -2; 276 0 -2 -2; 277 0.25 -2 -2; 280 0.5 -2 -2
    25. 282 -0.75 -2 -1.75; 283 -0.5 -2 -1.75; 284 -0.25 -2 -1.75; 285 0 -2 -1.75
    26. 286 0.25 -2 -1.75; 289 0.5 -2 -1.75; 291 -0.75 -2 -1.5; 292 -0.5 -2 -1.5
    27. 293 -0.25 -2 -1.5; 294 0 -2 -1.5; 295 0.25 -2 -1.5; 298 0.5 -2 -1.5
    28. 299 -0.25 0 -2.25; 300 -0.25 0 -0.75; 301 -1 -2 -0.25; 302 -1 -2 0
    29. 303 -1 -2 -0.5; 304 -1 -2 -0.75; 305 -1 -2 -2.5; 306 -1 -2 -2.25
    30. 307 -1 -2 -2.75; 308 -1 -2 -3; 309 -1 -2 -1; 310 -1 -2 -1.25; 311 -1 -2 -1.5
    31. 312 -1 -2 -2; 313 -1 -2 -1.75
    32. MEMBER INCIDENCES
    33. 193 215 299; 194 148 300
    34. ELEMENT INCIDENCES SHELL
    35. 50 124 123 122 121; 51 121 122 126 125; 55 131 130 123 124; 57 125 126 134 133
    36. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
    37. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
    38. 89 173 174 130 131; 90 174 175 139 130; 91 175 176 150 139; 100 190 186 23 189
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39. 101 189 23 187 191; 102 193 192 186 190; 104 191 187 196 195
40. 110 204 191 195 203; 111 205 189 191 204; 112 206 190 189 205
41. 113 207 193 190 206; 118 214 204 203 213; 119 215 205 204 214
42. 120 216 206 205 215; 121 217 207 206 216; 134 233 232 207 217
43. 135 232 234 193 207; 136 234 235 192 193; 146 147 247 246 146
44. 147 148 248 247 147; 148 149 249 248 148; 149 150 250 249 149
45. 152 176 253 250 150; 154 247 256 255 246; 155 248 257 256 247
46. 156 249 258 257 248; 157 250 259 258 249; 160 253 262 259 250
47. 162 256 292 291 255; 163 257 293 292 256; 164 258 294 293 257
48. 165 259 295 294 258; 168 262 298 295 259; 170 274 214 213 273
49. 171 275 215 214 274; 172 276 216 215 275; 173 277 217 216 276
50. 176 280 233 217 277; 178 283 274 273 282; 179 284 275 274 283
51. 180 285 276 275 284; 181 286 277 276 285; 184 289 280 277 286
52. 186 292 283 282 291; 187 293 284 283 292; 188 294 285 284 293
53. 189 295 286 285 294; 192 298 289 286 295; 195 133 134 301 302
54. 196 134 140 303 301; 197 140 146 304 303; 198 213 203 305 306
55. 199 203 195 307 305; 200 195 196 308 307; 201 146 246 309 304
56. 202 246 255 310 309; 203 255 291 311 310; 204 273 213 306 312
57. 205 282 273 312 313; 206 291 282 313 311
58. START GROUP DEFINITION
59. FLOOR
60. ELEMENT
61. _FDN_5_MOMENT 50 51 55 57 TO 61 66 69 89 TO 91 100 TO 102 104 110 TO 113 118 -
62. 121 134 TO 136 146 149 152 154 TO 157 160 162 TO 165 168 170 173 176 178 -
63. 179 TO 181 184 186 TO 189 192 195 TO 206
64. _FDN_5_SHEAR 50 51 55 57 89 TO 91 100 TO 102 104 134 TO 136 152 160 -
65. 162 TO 165 168 176 184 186 TO 189 192 195 TO 206
66. END GROUP DEFINITION
67. ELEMENT PROPERTY
68. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
69. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
70. 179 TO 181 184 186 TO 189 192 195 TO 206 THICKNESS 0.35
71. DEFINE MATERIAL START
72. ISOTROPIC CONCRETE
73. E 2.17185E+007
74. POISSON 0.17
75. DENSITY 25
76. ALPHA 1E-005
77. DAMP 0.05
78. TYPE CONCRETE
79. STRENGTH FCU 27579
80. END DEFINE MATERIAL
81. MEMBER PROPERTY AMERICAN
82. 193 194 PRIS YD 0.5 ZD 0.5
83. CONSTANTS
84. MATERIAL CONCRETE ALL
85. SUPPORTS
86. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
87. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
88. 179 TO 181 184 186 TO 189 192 195 TO 205 -
89. 206 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
```

JOINT	INFLUENCE AREA
23	0.031250

121	0.031250
122	0.062500
123	0.062500
124	0.031250
125	0.031250
126	0.062500
130	0.062500
131	0.031250
133	0.031250
134	0.062500
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062500
173	0.015625
174	0.031250
175	0.031250
176	0.031250
186	0.031250
187	0.031250
189	0.062500
190	0.062500
191	0.062500
192	0.031250
193	0.062500
195	0.062500
196	0.031250
203	0.062500
204	0.062500
205	0.062500
206	0.062500
207	0.062500
213	0.062500
214	0.062500
215	0.062500
216	0.062500
217	0.062500
232	0.031250
233	0.031250
234	0.031250
235	0.015625
246	0.062500
247	0.062500
248	0.062500
249	0.062500
250	0.062500
253	0.031250
255	0.062500
256	0.062500
257	0.062500

258	0.062500
259	0.062500
262	0.031250
273	0.062500
274	0.062500
275	0.062500
276	0.062500
277	0.062500
280	0.031250
282	0.062500
283	0.062500
284	0.062500
285	0.062500
286	0.062500
289	0.031250
291	0.062500
292	0.062500
293	0.062500
294	0.062500
295	0.062500
298	0.031250
301	0.031250
302	0.015625
303	0.031250
304	0.031250
305	0.031250
306	0.031250
307	0.031250
308	0.015625
309	0.031250
310	0.031250
311	0.031250
312	0.031250
313	0.031250
TOTAL	4.500000

```
 90. ***
 91. ************************************************
 92. ***LOAD COMBINATION FOR STRESS DESIGN
 93. ************************************************
 94. ***
 95. ***
 96. LOAD 1 LOADTYPE DEAD  TITLE DL
 97. ***
 98. SELFWEIGHT Y -1.1
 99. ***FROM GALLERY-1
100. JOINT LOAD
101. 299 FX 0 FY -0.942 FZ 0.007
102. 300 FX 0 FY -0.942 FZ -0.007
103. ELEMENT LOAD
104. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
105. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
106. 179 TO 181 184 186 TO 189 192 195 TO 206 PR GY -23
107. *
108. LOAD 2 LOADTYPE LIVE  TITLE LL
109. JOINT LOAD
110. 299 FX 4 FY -15.847 FZ 4.646
```

```
111. 300 FX 4 FY -19.353 FZ 4.354
112. *
113. LOAD 3 LOADTYPE LIVE  TITLE LL1
114. JOINT LOAD
115. 299 FX 4 FY -19.353 FZ -4.354
116. 300 FX 4 FY -15.847 FZ -4.646
117. *
118. LOAD 11 LOADTYPE WIND  TITLE WL+Z
119. JOINT LOAD
120. 299 FX 0 FY 0.72 FZ 1.375
121. 300 FX 0 FY -0.72 FZ 1.375
122. *
123. LOAD 12 LOADTYPE WIND  TITLE WL-Z
124. JOINT LOAD
125. 299 FX 0 FY -0.72 FZ -1.375
126. 300 FX 0 FY 0.72 FZ -1.375
127. *
128. LOAD 13 LOADTYPE WIND  TITLE WL+X
129. JOINT LOAD
130. 299 FX 1.375 FY 0 FZ 0
131. 300 FX 1.375 FY 0 FZ 0
132. *
133. LOAD 14 LOADTYPE WIND  TITLE WL-X
134. JOINT LOAD
135. 299 FX -1.375 FY 0 FZ 0
136. 300 FX -1.375 FY 0 FZ 0
137. *
138. LOAD 20 LOADTYPE NONE  TITLE TL1
139. JOINT LOAD
140. 299 FX 0 FY 0 FZ -19.506
141. 300 FX 0 FY 0 FZ 19.506
142. *
143. LOAD 21 LOADTYPE NONE  TITLE TL2
144. JOINT LOAD
145. 299 FX 0 FY 0 FZ 19.506
146. 300 FX 0 FY 0 FZ -19.506
147. *
148. LOAD 30 EARTHQUAKE LOAD (+Z)
149. JOINT LOAD
150. 299 FX 0 FY 0.072 FZ 0.139
151. 300 FX 0 FY -0.072 FZ 0.139
152. *
153. LOAD 31 EARTHQUAKE LOAD (-Z)
154. JOINT LOAD
155. 299 FX 0 FY -0.072 FZ -0.139
156. 300 FX 0 FY 0.072 FZ -0.139
157. *
158. LOAD 32 EARTHQUAKE LOAD (+X)
159. JOINT LOAD
160. 299 FX 0.139 FY 0 FZ 0
161. 300 FX 0.139 FY 0 FZ 0
162. *
163. LOAD 33 EARTHQUAKE LOAD (-X)
164. JOINT LOAD
165. 299 FX -0.139 FY 0 FZ 0
166. 300 FX -0.139 FY 0 FZ 0
```

```
167. *
168. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
169. *
170. LOAD COMB 201 D + L + TL1
171. 1 1.0 2 1.0 20 1.0
172. *
173. LOAD COMB 202 D + L1 + TL1
174. 1 1.0 3 1.0 20 1.0
175. *
176. LOAD COMB 203 D + L + TL2
177. 1 1.0 2 1.0 21 1.0
178. *
179. LOAD COMB 204 D + L1 + TL2
180. 1 1.0 3 1.0 21 1.0
181. *
182. LOAD COMB 205 D + 0.75XL + 0.75XTL1
183. 1 1.0 2 0.75 20 0.75
184. *
185. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
186. 1 1.0 3 0.75 20 0.75
187. *
188. LOAD COMB 207 D + 0.75XL + 0.75XTL2
189. 1 1.0 2 0.75 21 0.75
190. *
191. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
192. 1 1.0 3 0.75 21 0.75
193. *
194. LOAD COMB 209 D + WL(+Z)
195. 1 1.0 11 1.0
196. *
197. LOAD COMB 210 D + WL(+X)
198. 1 1.0 13 1.0
199. *
200. LOAD COMB 211 D + WL(-Z)
201. 1 1.0 12 1.0
202. *
203. LOAD COMB 212 D + WL(-X)
204. 1 1.0 14 1.0
205. *
206. LOAD COMB 213 D + 0.7XEL(+Z)
207. 1 1.0 30 0.7
208. *
209. LOAD COMB 214 D + 0.7XEL(+X)
210. 1 1.0 32 0.7
211. *
212. LOAD COMB 215 D + 0.7XEL(-Z)
213. 1 1.0 31 0.7
214. *
215. LOAD COMB 216 D + 0.7XEL(-X)
216. 1 1.0 33 0.7
217. *
218. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
219. 1 1.0 2 0.75 11 0.75
220. *
221. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
222. 1 1.0 3 0.75 11 0.75
```

```
223. *
224. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
225. 1 1.0 2 0.75 12 0.75
226. *
227. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
228. 1 1.0 3 0.75 12 0.75
229. *
230. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
231. 1 1.0 2 0.75 13 0.75
232. *
233. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
234. 1 1.0 3 0.75 13 0.75
235. *
236. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
237. 1 1.0 2 0.75 14 0.75
238. *
239. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
240. 1 1.0 3 0.75 14 0.75
241. *
242. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
243. 1 1.0 2 0.75 30 0.525
244. *
245. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
246. 1 1.0 3 0.75 30 0.525
247. *
248. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
249. 1 1.0 2 0.75 32 0.525
250. *
251. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
252. 1 1.0 3 0.75 32 0.525
253. *
254. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
255. 1 1.0 2 0.75 31 0.525
256. *
257. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
258. 1 1.0 3 0.75 31 0.525
259. *
260. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
261. 1 1.0 2 0.75 33 0.525
262. *
263. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
264. 1 1.0 3 0.75 33 0.525
265. *
266. LOAD COMB 233 0.6XD + WL(+Z)
267. 1 0.6 11 1.0
268. *
269. LOAD COMB 234 0.6XD + WL(+X)
270. 1 0.6 13 1.0
271. *
272. LOAD COMB 235 0.6XD + WL(-Z)
273. 1 0.6 12 1.0
274. *
275. LOAD COMB 236 0.6XD + WL(-X)
276. 1 0.6 14 1.0
277. *
278. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
```

```
279. 1 0.6 30 0.7
280. *
281. LOAD COMB 238 0.6XD + 0.7XEL(+X)
282. 1 0.6 32 0.7
283. *
284. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
285. 1 0.6 31 0.7
286. *
287. LOAD COMB 240 0.6XD + 0.7XEL(-X)
288. 1 0.6 33 0.7
289. *
290. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**93**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**72**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**91**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 72/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 285
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 13 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.6/ 63167.2 MB

```
291. CHANGE
292. *
293. *
294. *
295. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
296. LOAD COMB 100 1.4XD
297. 1 1.4
298. *
299. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
300. 1 1.2 20 1.2 2 1.6
301. *
302. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
303. 1 1.2 20 1.2 3 1.6
304. *
305. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
306. 1 1.2 21 1.2 2 1.6
307. *
308. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
309. 1 1.2 21 1.2 3 1.6
310. *
```

```
311. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
312. 1 1.2 11 0.8
313. *
314. LOAD COMB 106 1.2XD + 0.8XWL(+X)
315. 1 1.2 13 0.8
316. *
317. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
318. 1 1.2 12 0.8
319. *
320. LOAD COMB 108 1.2XD + 0.8XWL(-X)
321. 1 1.2 14 0.8
322. *
323. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
324. 1 0.9 11 1.6
325. *
326. LOAD COMB 110 0.9XD + 1.6XWL(+X)
327. 1 0.9 13 1.6
328. *
329. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
330. 1 0.9 12 1.6
331. *
332. LOAD COMB 112 0.9XD + 1.6XWL(-X)
333. 1 0.9 14 1.6
334. *
335. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
336. 1 1.2 2 0.5 1.6
337. *
338. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
339. 1 1.2 2 0.5 1.6
340. *
341. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
342. 1 1.2 2 0.5 1.6
343. *
344. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
345. 1 1.2 2 0.5 1.6
346. *
347. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
348. 1 1.2 3 0.5 1.6
349. *
350. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
351. 1 1.2 3 0.5 1.6
352. *
353. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
354. 1 1.2 3 0.5 1.6
355. *
356. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
357. 1 1.2 3 0.5 1.6
358. *
359. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
360. 1 1.2 2 0.5 1
361. *
362. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
363. 1 1.2 2 0.5 1
364. *
365. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
366. 1 1.2 2 0.5 1
```

```
367. *
368. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
369. 1 1.2 2 0.5 1
370. *
371. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
372. 1 1.2 3 0.5 1
373. *
374. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
375. 1 1.2 3 0.5 1
376. *
377. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
378. 1 1.2 3 0.5 1
379. *
380. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
381. 1 1.2 3 0.5 1
382. *
383. LOAD COMB 129 0.9XD + EL(+Z)
384. 1 0.9 30 1.0
385. *
386. LOAD COMB 130 0.9XD + EL(+X)
387. 1 0.9 32 1.0
388. *
389. LOAD COMB 131 0.9XD + EL(-Z)
390. 1 0.9 31 1.0
391. *
392. LOAD COMB 132 0.9XD + EL(-X)
393. 1 0.9 33 1.0
394. PERFORM ANALYSIS
395. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 19,2020 TIME= 15:32:47 ****


```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    OCT 19, 2020                 *
*           Time=    15:34:16                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\PROJECT FILE\MA'ADEN GOLD\Pipe Bridge\FOUNDATION\GPS-6 _7\WORKING\GPS-7 F7.STD
     2. START JOB INFORMATION
     3. *FOUNDATION F5
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 23 -0.25 -2 -3; 121 -0.25 -2 -0.15; 122 -0.25 -2 -0.4; 123 0 -2 -0.4
    10. 124 0 -2 -0.15; 125 -0.5 -2 -0.15; 126 -0.5 -2 -0.4; 130 0.25 -2 -0.4
    11. 131 0.25 -2 -0.15; 133 -0.75 -2 -0.15; 134 -0.75 -2 -0.4; 136 -0.25 -2 -0.65
    12. 137 0 -2 -0.65; 138 -0.5 -2 -0.65; 139 0.25 -2 -0.65; 140 -0.75 -2 -0.65
    13. 146 -0.75 -2 -0.9; 147 -0.5 -2 -0.9; 148 -0.25 -2 -0.9; 149 0 -2 -0.9
    14. 150 0.25 -2 -0.9; 173 0.5 -2 -0.15; 174 0.5 -2 -0.4; 175 0.5 -2 -0.65
    15. 176 0.5 -2 -0.9; 186 0 -2 -3; 187 -0.5 -2 -3; 189 -0.25 -2 -2.75
    16. 190 0 -2 -2.75; 191 -0.5 -2 -2.75; 192 0.25 -2 -3; 193 0.25 -2 -2.75
    17. 195 -0.75 -2 -2.75; 196 -0.75 -2 -3; 203 -0.75 -2 -2.5; 204 -0.5 -2 -2.5
    18. 205 -0.25 -2 -2.5; 206 0 -2 -2.5; 207 0.25 -2 -2.5; 213 -0.75 -2 -2.25
    19. 214 -0.5 -2 -2.25; 215 -0.25 -2 -2.25; 216 0 -2 -2.25; 217 0.25 -2 -2.25
    20. 232 0.5 -2 -2.5; 233 0.5 -2 -2.25; 234 0.5 -2 -2.75; 235 0.5 -2 -3
    21. 246 -0.75 -2 -1.15; 247 -0.5 -2 -1.15; 248 -0.25 -2 -1.15; 249 0 -2 -1.15
    22. 250 0.25 -2 -1.15; 253 0.5 -2 -1.15; 255 -0.75 -2 -1.4; 256 -0.5 -2 -1.4
    23. 257 -0.25 -2 -1.4; 258 0 -2 -1.4; 259 0.25 -2 -1.4; 262 0.5 -2 -1.4
    24. 273 -0.75 -2 -2; 274 -0.5 -2 -2; 275 -0.25 -2 -2; 276 0 -2 -2; 277 0.25 -2 -2
    25. 280 0.5 -2 -2; 282 -0.75 -2 -1.75; 283 -0.5 -2 -1.75; 284 -0.25 -2 -1.75
    26. 285 0 -2 -1.75; 286 0.25 -2 -1.75; 289 0.5 -2 -1.75; 291 -0.75 -2 -1.575
    27. 292 -0.5 -2 -1.575; 293 -0.25 -2 -1.575; 294 0 -2 -1.575; 295 0.25 -2 -1.575
    28. 298 0.5 -2 -1.575; 299 -0.25 0 -2.25; 300 -0.25 0 -0.9; 301 -1 -2 -0.4
    29. 302 -1 -2 -0.15; 303 -1 -2 -0.65; 304 -1 -2 -0.9; 305 -1 -2 -2.5
    30. 306 -1 -2 -2.25; 307 -1 -2 -2.75; 308 -1 -2 -3; 309 -1 -2 -1.15
    31. 310 -1 -2 -1.4; 311 -1 -2 -1.575; 312 -1 -2 -2; 313 -1 -2 -1.75
    32. MEMBER INCIDENCES
    33. 193 215 299; 194 148 300
    34. ELEMENT INCIDENCES SHELL
    35. 50 124 123 122 121; 51 121 122 126 125; 55 131 130 123 124; 57 125 126 134 133
    36. 58 126 138 140 134; 59 122 136 138 126; 60 123 137 136 122; 61 130 139 137 123
    37. 66 138 147 146 140; 67 136 148 147 138; 68 137 149 148 136; 69 139 150 149 137
    38. 89 173 174 130 131; 90 174 175 139 130; 91 175 176 150 139; 100 190 186 23 189
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39. 101 189 23 187 191; 102 193 192 186 190; 104 191 187 196 195
40. 110 204 191 195 203; 111 205 189 191 204; 112 206 190 189 205
41. 113 207 193 190 206; 118 214 204 203 213; 119 215 205 204 214
42. 120 216 206 205 215; 121 217 207 206 216; 134 233 232 207 217
43. 135 232 234 193 207; 136 234 235 192 193; 146 147 247 246 146
44. 147 148 248 247 147; 148 149 249 248 148; 149 150 250 249 149
45. 152 176 253 250 150; 154 247 256 255 246; 155 248 257 256 247
46. 156 249 258 257 248; 157 250 259 258 249; 160 253 262 259 250
47. 162 256 292 291 255; 163 257 293 292 256; 164 258 294 293 257
48. 165 259 295 294 258; 168 262 298 295 259; 170 274 214 213 273
49. 171 275 215 214 274; 172 276 216 215 275; 173 277 217 216 276
50. 176 280 233 217 277; 178 283 274 273 282; 179 284 275 274 283
51. 180 285 276 275 284; 181 286 277 276 285; 184 289 280 277 286
52. 186 292 283 282 291; 187 293 284 283 292; 188 294 285 284 293
53. 189 295 286 285 294; 192 298 289 286 295; 195 133 134 301 302
54. 196 134 140 303 301; 197 140 146 304 303; 198 213 203 305 306
55. 199 203 195 307 305; 200 195 196 308 307; 201 146 246 309 304
56. 202 246 255 310 309; 203 255 291 311 310; 204 273 213 306 312
57. 205 282 273 312 313; 206 291 282 313 311
58. START GROUP DEFINITION
59. FLOOR
60. ELEMENT
61. _FDN_5_MOMENT 50 51 55 57 TO 61 66 69 89 TO 91 100 TO 102 104 110 TO 113 118 -
62. 121 134 TO 136 146 149 152 154 TO 157 160 162 TO 165 168 170 173 176 178 -
63. 179 TO 181 184 186 TO 189 192 195 TO 206
64. _FDN_5_SHEAR 50 51 55 57 89 TO 91 100 TO 102 104 134 TO 136 152 160 -
65. 162 TO 165 168 176 184 186 TO 189 192 195 TO 206
66. END GROUP DEFINITION
67. ELEMENT PROPERTY
68. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
69. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
70. 179 TO 181 184 186 TO 189 192 195 TO 206 THICKNESS 0.35
71. DEFINE MATERIAL START
72. ISOTROPIC CONCRETE
73. E 2.17185E+007
74. POISSON 0.17
75. DENSITY 25
76. ALPHA 1E-005
77. DAMP 0.05
78. TYPE CONCRETE
79. STRENGTH FCU 27579
80. END DEFINE MATERIAL
81. MEMBER PROPERTY AMERICAN
82. 193 194 PRIS YD 0.5 ZD 0.5
83. CONSTANTS
84. MATERIAL CONCRETE ALL
85. SUPPORTS
86. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
87. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
88. 179 TO 181 184 186 TO 189 192 195 TO 205 -
89. 206 PLATE MAT DIRECT Y SUBGRADE 54000 PRINT
```

JOINT	INFLUENCE AREA
23	0.031250

121	0.031250
122	0.062500
123	0.062500
124	0.031250
125	0.031250
126	0.062500
130	0.062500
131	0.031250
133	0.031250
134	0.062500
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062500
173	0.015625
174	0.031250
175	0.031250
176	0.031250
186	0.031250
187	0.031250
189	0.062500
190	0.062500
191	0.062500
192	0.031250
193	0.062500
195	0.062500
196	0.031250
203	0.062500
204	0.062500
205	0.062500
206	0.062500
207	0.062500
213	0.062500
214	0.062500
215	0.062500
216	0.062500
217	0.062500
232	0.031250
233	0.031250
234	0.031250
235	0.015625
246	0.062500
247	0.062500
248	0.062500
249	0.062500
250	0.062500
253	0.031250
255	0.053125
256	0.053125
257	0.053125

258	0.053125
259	0.053125
262	0.026563
273	0.062500
274	0.062500
275	0.062500
276	0.062500
277	0.062500
280	0.031250
282	0.053125
283	0.053125
284	0.053125
285	0.053125
286	0.053125
289	0.026563
291	0.043750
292	0.043750
293	0.043750
294	0.043750
295	0.043750
298	0.021875
301	0.031250
302	0.015625
303	0.031250
304	0.031250
305	0.031250
306	0.031250
307	0.031250
308	0.015625
309	0.031250
310	0.026563
311	0.021875
312	0.031250
313	0.026563
TOTAL	4.274999

```
 90. ***
 91. **************************************************
 92. ***LOAD COMBINATION FOR STRESS DESIGN
 93. **************************************************
 94. ***
 95. ***
 96. LOAD 1 LOADTYPE DEAD  TITLE DL
 97. ***
 98. SELFWEIGHT Y -1.1
 99. ***FROM GALLERY-1
100. JOINT LOAD
101. 299 FX 0 FY -0.942 FZ 0.007
102. 300 FX 0 FY -0.942 FZ -0.007
103. ELEMENT LOAD
104. 50 51 55 57 TO 61 66 TO 69 89 TO 91 100 TO 102 104 110 TO 113 118 TO 121 134 -
105. 135 TO 136 146 TO 149 152 154 TO 157 160 162 TO 165 168 170 TO 173 176 178 -
106. 179 TO 181 184 186 TO 189 192 195 TO 206 PR GY -23
107. *
108. LOAD 2 LOADTYPE LIVE  TITLE LL
109. JOINT LOAD
110. 299 FX 4 FY -15.847 FZ 4.646
```

```
111. 300 FX 4 FY -19.353 FZ 4.354
112. *
113. LOAD 3 LOADTYPE LIVE  TITLE LL1
114. JOINT LOAD
115. 299 FX 4 FY -19.353 FZ -4.354
116. 300 FX 4 FY -15.847 FZ -4.646
117. *
118. LOAD 11 LOADTYPE WIND  TITLE WL+Z
119. JOINT LOAD
120. 299 FX 0 FY 0.72 FZ 1.375
121. 300 FX 0 FY -0.72 FZ 1.375
122. *
123. LOAD 12 LOADTYPE WIND  TITLE WL-Z
124. JOINT LOAD
125. 299 FX 0 FY -0.72 FZ -1.375
126. 300 FX 0 FY 0.72 FZ -1.375
127. *
128. LOAD 13 LOADTYPE WIND  TITLE WL+X
129. JOINT LOAD
130. 299 FX 1.375 FY 0 FZ 0
131. 300 FX 1.375 FY 0 FZ 0
132. *
133. LOAD 14 LOADTYPE WIND  TITLE WL-X
134. JOINT LOAD
135. 299 FX -1.375 FY 0 FZ 0
136. 300 FX -1.375 FY 0 FZ 0
137. *
138. LOAD 20 LOADTYPE NONE  TITLE TL1
139. JOINT LOAD
140. 299 FX 0 FY 0 FZ -19.506
141. 300 FX 0 FY 0 FZ 19.506
142. *
143. LOAD 21 LOADTYPE NONE  TITLE TL2
144. JOINT LOAD
145. 299 FX 0 FY 0 FZ 19.506
146. 300 FX 0 FY 0 FZ -19.506
147. *
148. LOAD 30 EARTHQUAKE LOAD (+Z)
149. JOINT LOAD
150. 299 FX 0 FY 0.072 FZ 0.139
151. 300 FX 0 FY -0.072 FZ 0.139
152. *
153. LOAD 31 EARTHQUAKE LOAD (-Z)
154. JOINT LOAD
155. 299 FX 0 FY -0.072 FZ -0.139
156. 300 FX 0 FY 0.072 FZ -0.139
157. *
158. LOAD 32 EARTHQUAKE LOAD (+X)
159. JOINT LOAD
160. 299 FX 0.139 FY 0 FZ 0
161. 300 FX 0.139 FY 0 FZ 0
162. *
163. LOAD 33 EARTHQUAKE LOAD (-X)
164. JOINT LOAD
165. 299 FX -0.139 FY 0 FZ 0
166. 300 FX -0.139 FY 0 FZ 0
```

```
167. *
168. ****** LOAD COMBINATIONS FOR SERVICEABILITY CHECK
169. *
170. LOAD COMB 201 D + L + TL1
171. 1 1.0 2 1.0 20 1.0
172. *
173. LOAD COMB 202 D + L1 + TL1
174. 1 1.0 3 1.0 20 1.0
175. *
176. LOAD COMB 203 D + L + TL2
177. 1 1.0 2 1.0 21 1.0
178. *
179. LOAD COMB 204 D + L1 + TL2
180. 1 1.0 3 1.0 21 1.0
181. *
182. LOAD COMB 205 D + 0.75XL + 0.75XTL1
183. 1 1.0 2 0.75 20 0.75
184. *
185. LOAD COMB 206 D + 0.75XL1 + 0.75XTL1
186. 1 1.0 3 0.75 20 0.75
187. *
188. LOAD COMB 207 D + 0.75XL + 0.75XTL2
189. 1 1.0 2 0.75 21 0.75
190. *
191. LOAD COMB 208 D + 0.75XL1 + 0.75XTL2
192. 1 1.0 3 0.75 21 0.75
193. *
194. LOAD COMB 209 D + WL(+Z)
195. 1 1.0 11 1.0
196. *
197. LOAD COMB 210 D + WL(+X)
198. 1 1.0 13 1.0
199. *
200. LOAD COMB 211 D + WL(-Z)
201. 1 1.0 12 1.0
202. *
203. LOAD COMB 212 D + WL(-X)
204. 1 1.0 14 1.0
205. *
206. LOAD COMB 213 D + 0.7XEL(+Z)
207. 1 1.0 30 0.7
208. *
209. LOAD COMB 214 D + 0.7XEL(+X)
210. 1 1.0 32 0.7
211. *
212. LOAD COMB 215 D + 0.7XEL(-Z)
213. 1 1.0 31 0.7
214. *
215. LOAD COMB 216 D + 0.7XEL(-X)
216. 1 1.0 33 0.7
217. *
218. LOAD COMB 217 D + 0.75XL + 0.75XWL(+Z)
219. 1 1.0 2 0.75 11 0.75
220. *
221. LOAD COMB 218 D + 0.75XL1 + 0.75XWL(+Z)
222. 1 1.0 3 0.75 11 0.75
```

```
223. *
224. LOAD COMB 219 D + 0.75XL + 0.75XWL(-Z)
225. 1 1.0 2 0.75 12 0.75
226. *
227. LOAD COMB 220 D + 0.75XL1 + 0.75XWL(-Z)
228. 1 1.0 3 0.75 12 0.75
229. *
230. LOAD COMB 221 D + 0.75XL + 0.75XWL(+X)
231. 1 1.0 2 0.75 13 0.75
232. *
233. LOAD COMB 222 D + 0.75XL1 + 0.75XWL(+X)
234. 1 1.0 3 0.75 13 0.75
235. *
236. LOAD COMB 223 D + 0.75XL + 0.75XWL(-X)
237. 1 1.0 2 0.75 14 0.75
238. *
239. LOAD COMB 224 D + 0.75XL1 + 0.75XWL(-X)
240. 1 1.0 3 0.75 14 0.75
241. *
242. LOAD COMB 225 D + 0.75XL + 0.525XEL(+Z)
243. 1 1.0 2 0.75 30 0.525
244. *
245. LOAD COMB 226 D + 0.75XL1 + 0.525XEL(+Z)
246. 1 1.0 3 0.75 30 0.525
247. *
248. LOAD COMB 227 D + 0.75XL + 0.525XEL(+X)
249. 1 1.0 2 0.75 32 0.525
250. *
251. LOAD COMB 228 D + 0.75XL1 + 0.525XEL(+X)
252. 1 1.0 3 0.75 32 0.525
253. *
254. LOAD COMB 229 D + 0.75XL + 0.525XEL(-Z)
255. 1 1.0 2 0.75 31 0.525
256. *
257. LOAD COMB 230 D + 0.75XL1 + 0.525XEL(-Z)
258. 1 1.0 3 0.75 31 0.525
259. *
260. LOAD COMB 231 D + 0.75XL + 0.525XEL(-X)
261. 1 1.0 2 0.75 33 0.525
262. *
263. LOAD COMB 232 D + 0.75XL1 + 0.525XEL(-X)
264. 1 1.0 3 0.75 33 0.525
265. *
266. LOAD COMB 233 0.6XD + WL(+Z)
267. 1 0.6 11 1.0
268. *
269. LOAD COMB 234 0.6XD + WL(+X)
270. 1 0.6 13 1.0
271. *
272. LOAD COMB 235 0.6XD + WL(-Z)
273. 1 0.6 12 1.0
274. *
275. LOAD COMB 236 0.6XD + WL(-X)
276. 1 0.6 14 1.0
277. *
278. LOAD COMB 237 0.6XD + 0.7XEL(+Z)
```

```
279. 1 0.6 30 0.7
280. *
281. LOAD COMB 238 0.6XD + 0.7XEL(+X)
282. 1 0.6 32 0.7
283. *
284. LOAD COMB 239 0.6XD + 0.7XEL(-Z)
285. 1 0.6 31 0.7
286. *
287. LOAD COMB 240 0.6XD + 0.7XEL(-X)
288. 1 0.6 33 0.7
289. *
290. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**93**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**72**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**91**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 72/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 285
TOTAL LOAD COMBINATION CASES = 40 SO FAR.
SIZE OF STIFFNESS MATRIX = 13 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.6/ 63164.6 MB

```
291. CHANGE
292. *
293. *
294. *
295. ****** LOAD COMBINATIONS FOR STRENGTH DESIGN
296. LOAD COMB 100 1.4XD
297. 1 1.4
298. *
299. LOAD COMB 101 1.2XD + 1.2XTL1 + 1.6XL
300. 1 1.2 20 1.2 2 1.6
301. *
302. LOAD COMB 102 1.2XD + 1.2XTL1 + 1.6XL1
303. 1 1.2 20 1.2 3 1.6
304. *
305. LOAD COMB 103 1.2XD + 1.2XTL2 + 1.6XL
306. 1 1.2 21 1.2 2 1.6
307. *
308. LOAD COMB 104 1.2XD + 1.2XTL2 + 1.6XL1
309. 1 1.2 21 1.2 3 1.6
310. *
```

```
311. LOAD COMB 105 1.2XD + 0.8XWL(+Z)
312. 1 1.2 11 0.8
313. *
314. LOAD COMB 106 1.2XD + 0.8XWL(+X)
315. 1 1.2 13 0.8
316. *
317. LOAD COMB 107 1.2XD + 0.8XWL(-Z)
318. 1 1.2 12 0.8
319. *
320. LOAD COMB 108 1.2XD + 0.8XWL(-X)
321. 1 1.2 14 0.8
322. *
323. LOAD COMB 109 0.9XD + 1.6XWL(+Z)
324. 1 0.9 11 1.6
325. *
326. LOAD COMB 110 0.9XD + 1.6XWL(+X)
327. 1 0.9 13 1.6
328. *
329. LOAD COMB 111 0.9XD + 1.6XWL(-Z)
330. 1 0.9 12 1.6
331. *
332. LOAD COMB 112 0.9XD + 1.6XWL(-X)
333. 1 0.9 14 1.6
334. *
335. LOAD COMB 113 1.2XD + 0.5XL + 1.6XWL (+Z)
336. 1 1.2 2 0.5 1.6
337. *
338. LOAD COMB 114 1.2XD + 0.5XL + 1.6XWL (+X)
339. 1 1.2 2 0.5 1.6
340. *
341. LOAD COMB 115 1.2XD + 0.5XL + 1.6XWL (-Z)
342. 1 1.2 2 0.5 1.6
343. *
344. LOAD COMB 116 1.2XD + 0.5XL + 1.6XWL (-X)
345. 1 1.2 2 0.5 1.6
346. *
347. LOAD COMB 117 1.2XD + 0.5XL1 + 1.6XWL (+Z)
348. 1 1.2 3 0.5 1.6
349. *
350. LOAD COMB 118 1.2XD + 0.5XL1 + 1.6XWL (+X)
351. 1 1.2 3 0.5 1.6
352. *
353. LOAD COMB 119 1.2XD + 0.5XL1 + 1.6XWL (-Z)
354. 1 1.2 3 0.5 1.6
355. *
356. LOAD COMB 120 1.2XD + 0.5XL1 + 1.6XWL (-X)
357. 1 1.2 3 0.5 1.6
358. *
359. LOAD COMB 121 1.2XD + 0.5XL + EL (+Z)
360. 1 1.2 2 0.5 1
361. *
362. LOAD COMB 122 1.2XD + 0.5XL + EL (+X)
363. 1 1.2 2 0.5 1
364. *
365. LOAD COMB 123 1.2XD + 0.5XL + EL (-Z)
366. 1 1.2 2 0.5 1
```

```
367. *
368. LOAD COMB 124 1.2XD + 0.5XL + EL (-X)
369. 1 1.2 2 0.5 1
370. *
371. LOAD COMB 125 1.2XD + 0.5XL1 + EL (+Z)
372. 1 1.2 3 0.5 1
373. *
374. LOAD COMB 126 1.2XD + 0.5XL1 + EL (+X)
375. 1 1.2 3 0.5 1
376. *
377. LOAD COMB 127 1.2XD + 0.5XL1 + EL (-Z)
378. 1 1.2 3 0.5 1
379. *
380. LOAD COMB 128 1.2XD + 0.5XL1 + EL (-X)
381. 1 1.2 3 0.5 1
382. *
383. LOAD COMB 129 0.9XD + EL(+Z)
384. 1 0.9 30 1.0
385. *
386. LOAD COMB 130 0.9XD + EL(+X)
387. 1 0.9 32 1.0
388. *
389. LOAD COMB 131 0.9XD + EL(-Z)
390. 1 0.9 31 1.0
391. *
392. LOAD COMB 132 0.9XD + EL(-X)
393. 1 0.9 33 1.0
394. PERFORM ANALYSIS
395. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 19,2020 TIME= 15:34:19 ****

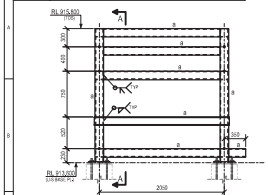

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
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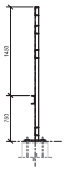
ATTACHMENT 2 SUPERSTRUCTURE DRAWINGS

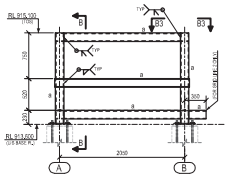
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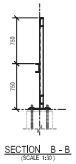
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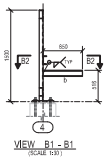
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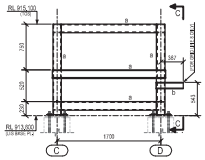


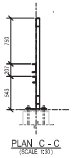


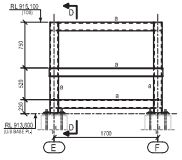


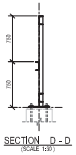


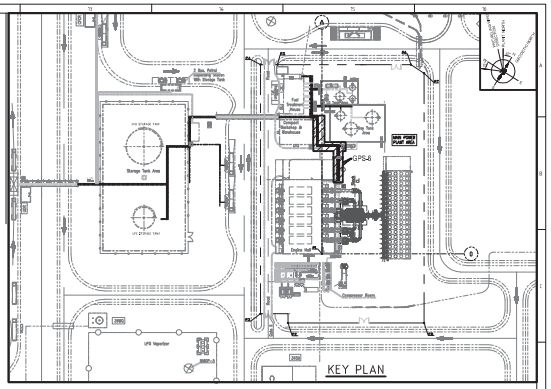


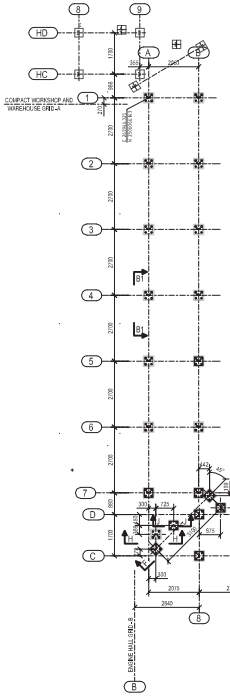


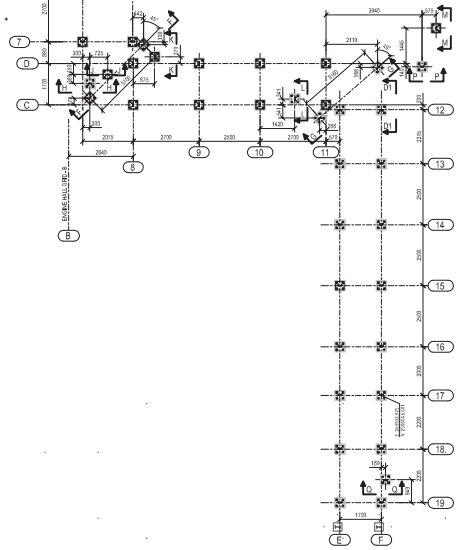






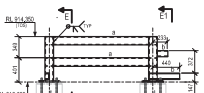


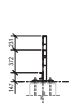


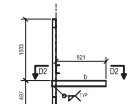


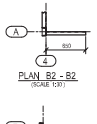
PLAN AT BASE PLATE LEVEL OF PIPE SUPPORT MARKED GPS-6
(SCALE 1:75)

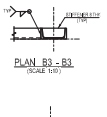


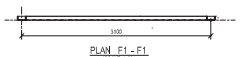


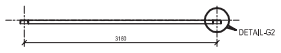


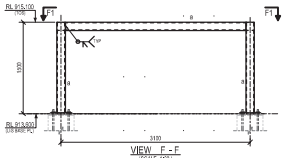


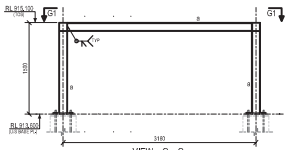


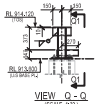


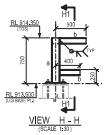


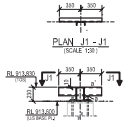


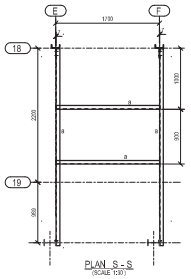


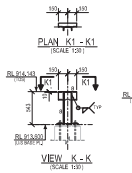


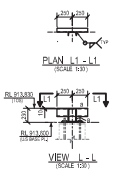


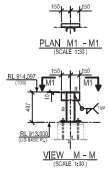


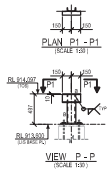


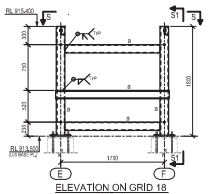


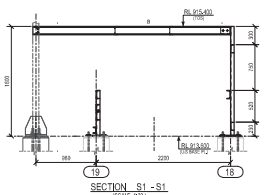


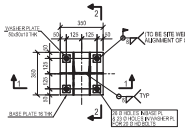


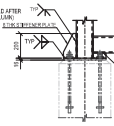


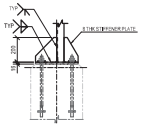












SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
a	MP4140	]	-	-	MATERIAL GRADE E275 JR
b	MP4100	]	-	-	

ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION, MINIMUM 2 NOS. M20 BOLTS SHALL BE PROVIDED.

LEGEND :-

RL = REDUCED LEVEL
TOS = TOP OF STEEL
TYP = TYPICAL
THK = THICK
U/S = UNDER SIDE
PL = PLATE
HD = HOLDING DOWN
ELEV= ELEVATION
ϕ = DIAMETER

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES & LEVELS ARE IN METRES UNLESS OTHERWISE SPECIFIED.
2. ALL WELDS SHALL BE OF SHOP QUALITY AND TO BE TESTED AS PER QUALITY ASSURANCE PLAN (QAP).

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	ENG	APP	TYP
MD-216-49PR-EG-SL-LAE-9503-002	LAYOUT POWER GENERATION PLAN	A01	09/10/2020	ISSUED FOR REFERENCE	SWB	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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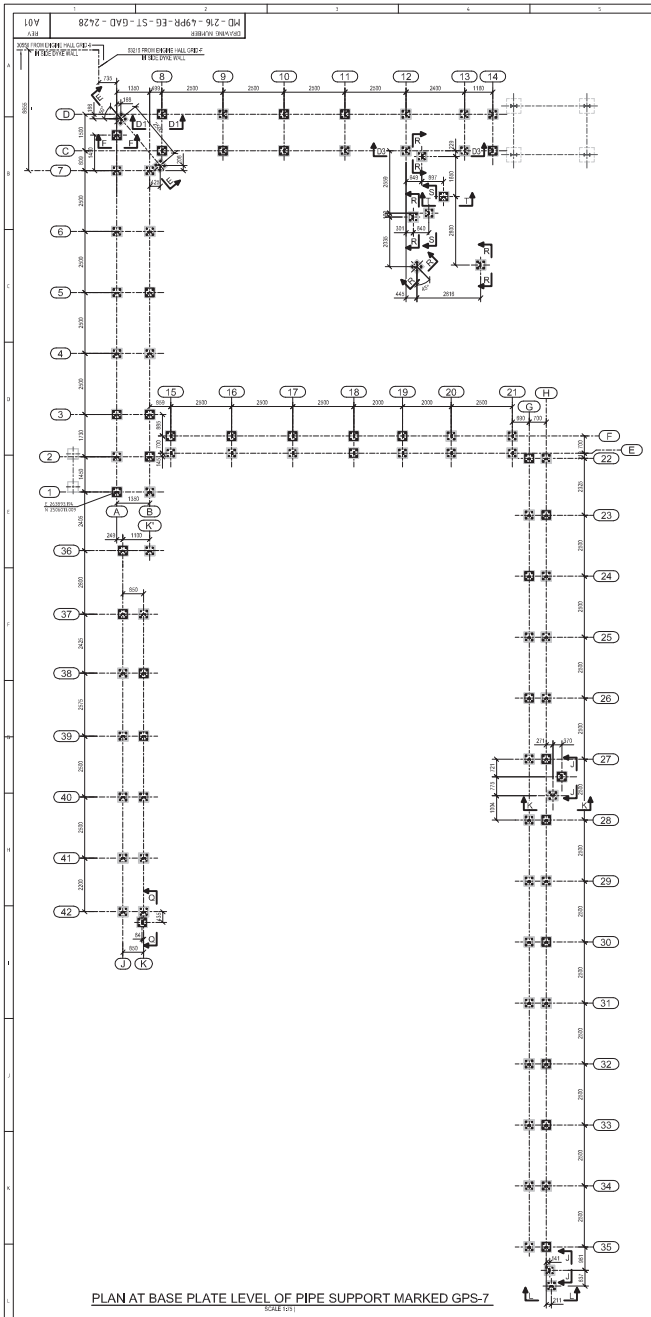
Outotec

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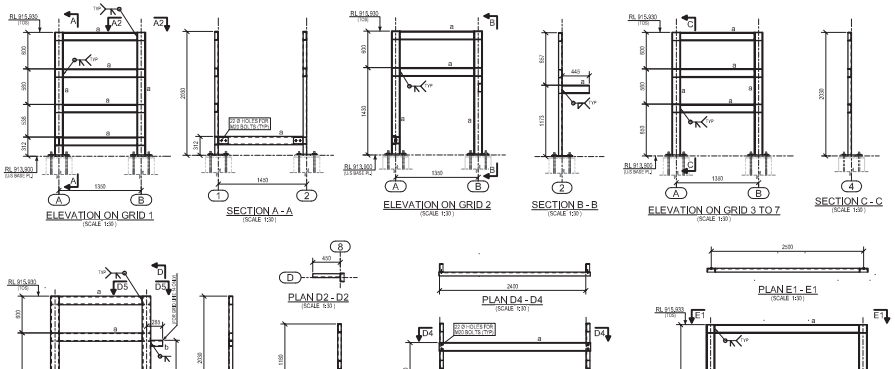
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_008_S_G_26109_010			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GA001__WG01_MTL01__45121		OU100697481	0
DRAWN	SWB	15/09/2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	UR	12/09/2020	GENERAL ARRANGEMENT
ENGINEER	SVK	16/09/2020	POWER PLANT PIPE RACK
APPROVED	AG	09/10/2020	PLAN, ELEV & DETAILS OF PIPE SUPPORT GPS-6
COMPLETION			
SCALE	1:75, 30, 15	A0	DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-2427 REV: A01

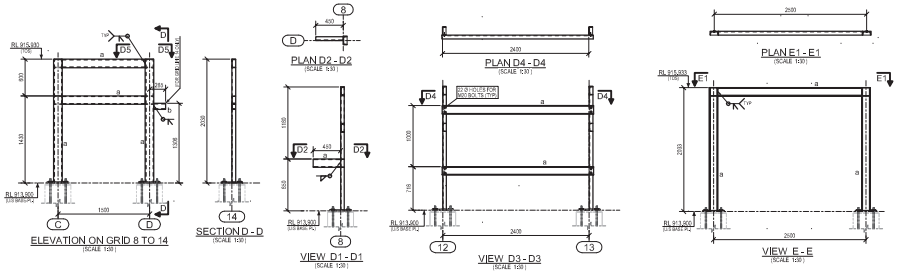


PLAN AT BASE PLATE LEVEL OF PIPE SUPPORT MARKED GPS-7

SCALE 1:75

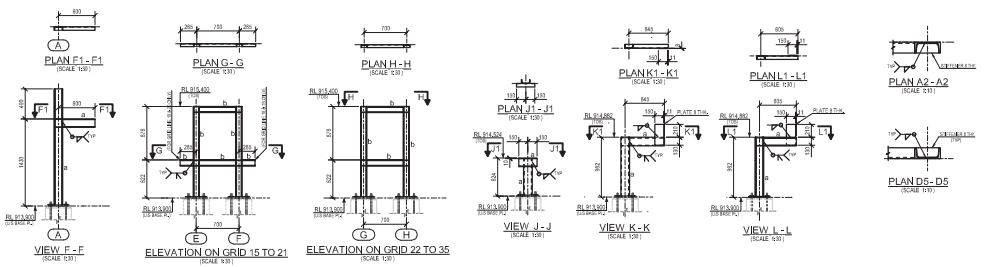


ELEVATION ON GRID 1 (SCALE 1:30) | SECTION A - A (SCALE 1:30) | ELEVATION ON GRID 2 (SCALE 1:30) | SECTION B - B (SCALE 1:30) | ELEVATION ON GRID 3 TO 7 (SCALE 1:30) | SECTION C - C (SCALE 1:30)



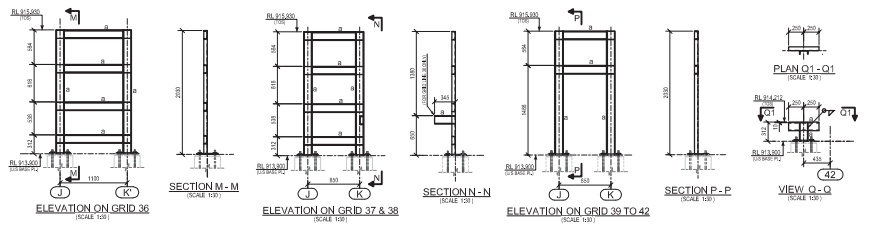
ELEVATION ON GRID 8 TO 14 (SCALE 1:30) | SECTION D - D (SCALE 1:30) | VIEW D1 - D1 (SCALE 1:30) | VIEW D3 - D3 (SCALE 1:30) | VIEW E - E (SCALE 1:30)

PLAN D2 - D2 (SCALE 1:30) | PLAN D4 - D4 (SCALE 1:30) | PLAN E1 - E1 (SCALE 1:30)

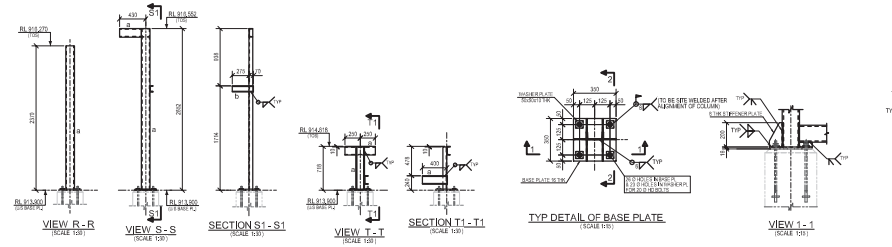


PLAN F1 - F1 (SCALE 1:30) | PLAN G - G (SCALE 1:30) | PLAN H - H (SCALE 1:30) | PLAN K1 - K1 (SCALE 1:30) | PLAN L1 - L1 (SCALE 1:30) | PLAN A2 - A2 (SCALE 1:10) | PLAN D5 - D5 (SCALE 1:10)

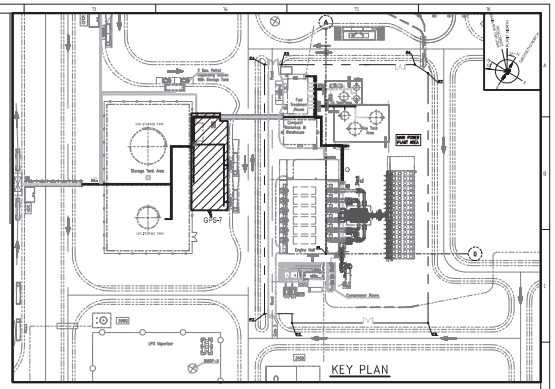
VIEW F - F (SCALE 1:30) | ELEVATION ON GRID 15 TO 21 (SCALE 1:30) | ELEVATION ON GRID 22 TO 35 (SCALE 1:30) | PLAN J1 - J1 (SCALE 1:30) | VIEW J - J (SCALE 1:30) | VIEW K - K (SCALE 1:30) | VIEW L - L (SCALE 1:30)



ELEVATION ON GRID 36 (SCALE 1:30) | SECTION M - M (SCALE 1:30) | ELEVATION ON GRID 37 & 38 (SCALE 1:30) | SECTION N - N (SCALE 1:30) | ELEVATION ON GRID 39 TO 42 (SCALE 1:30) | SECTION P - P (SCALE 1:30) | PLAN Q1 - Q1 (SCALE 1:30) | VIEW Q - Q (SCALE 1:30)



VIEW R - R (SCALE 1:30) | VIEW S - S (SCALE 1:30) | SECTION S1 - S1 (SCALE 1:30) | VIEW T - T (SCALE 1:30) | SECTION T1 - T1 (SCALE 1:30) | TYP DETAIL OF BASE PLATE (SCALE 1:15) | VIEW 1 - 1 (SCALE 1:20) | VIEW 2 - 2 (SCALE 1:30)



KEY PLAN

SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
a	UPN140	]	-	-	MATERIAL GRADE: IS2062 GR. E250
b	UPN100	]	-	-	

ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

LEGEND :-

RL = REDUCED LEVEL
TOS = TOP OF STEEL
TYP = TYPICAL
THK = THICK
U/S = UNDER SIDE
PL = PLATE
HD = HOLDING DOWN
ELEV= ELEVATION
ϕ = DIAMETER

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES & LEVELS ARE IN METRES UNLESS OTHERWISE SPECIFIED.
2. ALL WELDS SHALL BE OF SHOP QUALITY AND TO BE TESTED AS PER QUALITY ASSURANCE PLAN (QAP).

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	ENG	APP	TYP	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	DRW	CHK	ENG	APP	TYP
PD-216-49PR-EG-SL-LAC-9503-1002	LAYOUT POWER GENERATION PLAN	A01	09/10/2020	ISSUED FOR REFERENCE	SWB	UR	SVK	AG										

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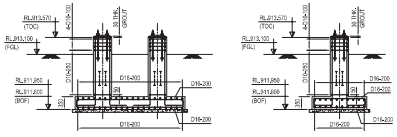
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DRAWN	SWB	09/09/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	UR	21/09/2020	GENERAL ARRANGEMENT
ENGINEER	SVK	09/09/2020	POWER PLANT PIPE RACK
APPROVED	AG	09/10/2020	PLAN, ELEV & DETAILS OF PIPE SUPPORT GPS-7
EMPLOYER			
SCALE	1:75, 30, 15	A0	DRAWING NUMBER: MD-216-49PR-EG-ST-GAD-2428 REV: A01

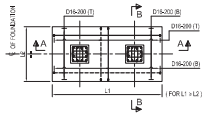
ATTACHMENT 3 CIVIL DRAWINGS

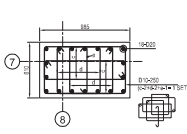
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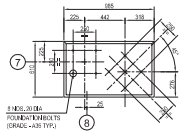
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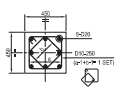
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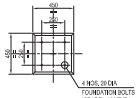


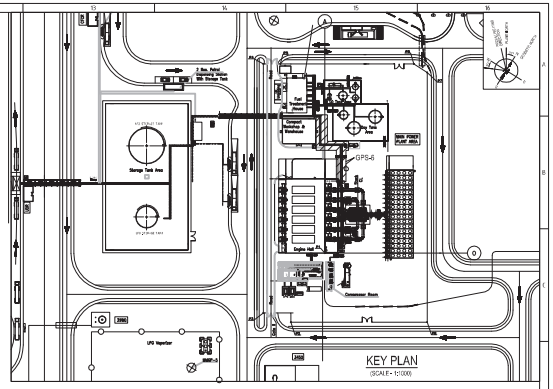


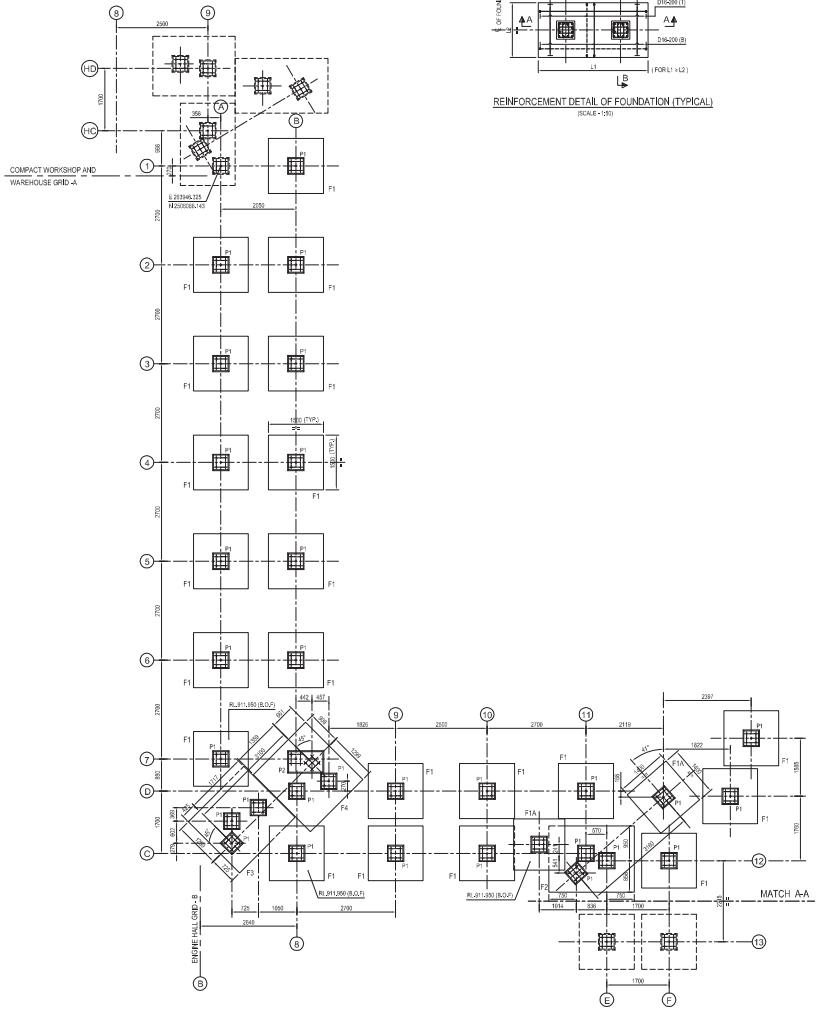












FOUNDATION LAYOUT PLAN AT RL.911,600 B,O,F. (U,N,O.)
(SCALE 1:50)

SCHEDULE OF BOLTS	
BOLT DIA. (MM)	BOLT NUMBERS REQUIRED
20	192

LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOUNDATION
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
THK = THICK
GA = GENERAL ARRANGEMENT
RC = REINFORCED CONCRETE
PCC = PLAIN CEMENT CONCRETE
DIA = DIAMETER
T = TOP
B = BOTTOM
RL = REDUCED LEVEL
UNO = UNLESS NOTED OTHERWISE
= TOP REINFORCEMENT
= BOTTOM REINFORCEMENT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATIONS & 13 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 45 KN/M SQ.M.
5. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
6. IF ANY PART OF FOUNDATION COMES ON BACK FILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. FOR ALL OTHER NOTES REFER DRAWING NO. 100-10-SIN0AEG-GEN-2001 TO 0005.
8. FOR GENERAL NOTES OF CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. 100-10-100AEGCIVGEN-2001 & 1005.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	MD-216-49PR-EG-ST-GAD-2027								
	DETAIL OF PIPE SUPPORT GPS-6	A01	15/10/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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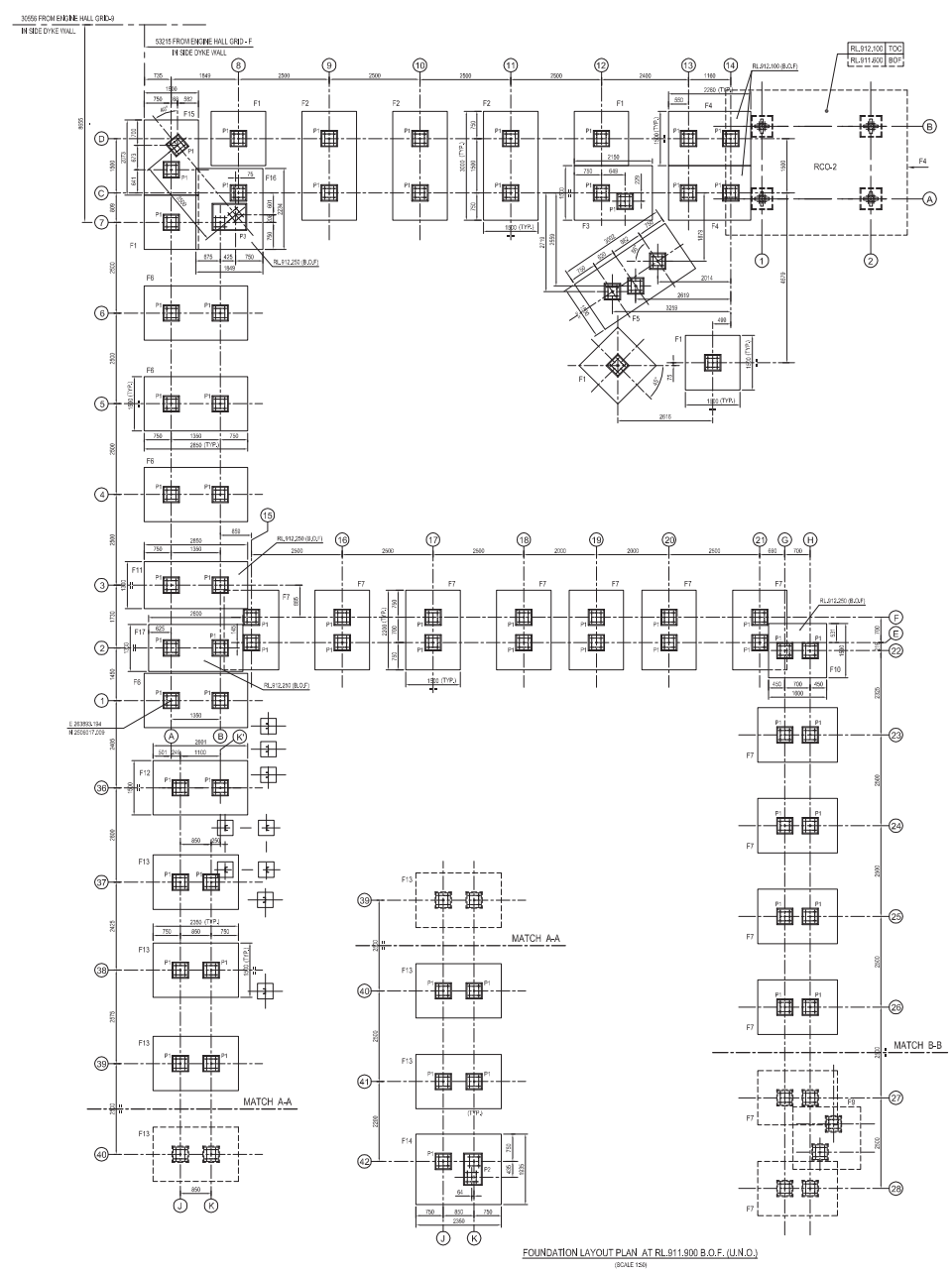
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OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
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DRAWN	ABH	05/10/2020
CHECKED	UR	13/10/2020
ENGINEER	SVK	05/10/2020
APPROVED	AG	15/10/2020
EMPLOYER		
SCALE	1:100	A0

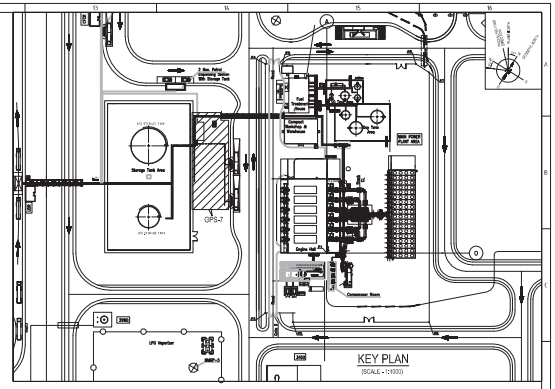
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT PIPE RACK
G.A. AND DETAIL OF FOUNDATION FOR GPS-6

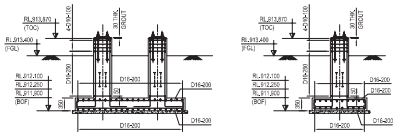
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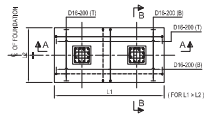
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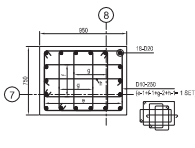


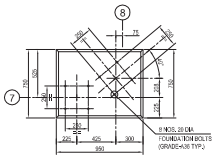
FOUNDATION LAYOUT PLAN AT RL.911.900 B.O.F. (U.N.O.)
(SCALE 1:50)

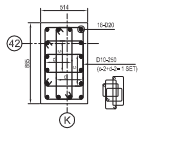


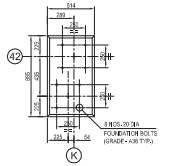


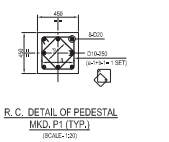


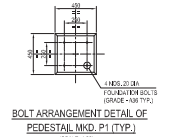












SCHEDULE OF BOLTS	
BOLT DIA. (MM)	BOLT NUMBERS REQUIRED
20	392

LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOUNDATION
TYP = TYPICAL
MKD = MARKED
℄ = CENTERLINE
THK = THICK
GA = GENERAL ARRANGEMENT
RC = REINFORCED CONCRETE
PCC = PLAIN CEMENT CONCRETE
DIA = DIAMETER
T = TOP
B = BOTTOM
RL = REDUCED LEVEL
UNO = UNLESS NOTED OTHERWISE
— — — = TOP REINFORCEMENT
——— = BOTTOM REINFORCEMENT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATIONS & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR-420 MPa (GRADE 60) - ASTM A615/615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 45 TON/SQ.M.
5. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. FOR ALL OTHER NOTES REFER DRAWING NO. X00-T1-SIMDAG-EG-S1-2001-2001 TO 0005.
8. FOR GENERAL NOTES OF CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. X00-T1-1004-EG-SADEG-2001 & 1005.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	MD-216-49PR-EG-ST-GAD-2028			DETAIL OF PIPE SUPPORT GPS-7					
		A01	15/10/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT PIPE RACK
G.A. AND DETAIL OF FOUNDATION FOR GPS-7

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